JOMO KENYATTA UNIVERSITY OF AGRICULTURE AND TECHNOLOGY

SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY

DEPARTMENT OF COMPUTING

WRAP MODELLING: WILDFIRE RISK AND PROGRESSION MODELLING

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REGISTRATION NUMBER: SCT212-0105/2022

COURSE: BSC COMPUTER TECHNOLOGY

UNIT: PROJECTS

UNIT CODE: BCT 2411

DATE OF SUBMISSION: 18/06/2026

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Abstract

Wildfire incidents across East Africa are intensifying in frequency and severity, yet the region remains critically underserved by existing fire spread modelling systems, which are predominantly calibrated for North American and Mediterranean ecosystems. This document presents WRaP (Wildfire Risk and Progression Modelling), a dual-phase fire intelligence system built upon a Rothermel Rate of Spread (ROS) model embedded within a Cellular Automata (CA) engine. The system is designed to better represent the Afromontane and tropical forest ecosystems found in Kenya and across East Africa, addressing a critical gap in regionally appropriate fire modelling tools.

The CA engine discretises the region of interest into a two-dimensional grid of 100-metre cells. Each cell is an automaton assignable to one of four states: UNBURNED, BURNING, BURNED, or NON-COMBUSTIBLE. Transition probabilities at each generation are computed by evaluating Moore neighbourhood interactions, where the Rate of Spread toward each unburned cell from any burning neighbour is derived from a simplified Rothermel (1972) surface fire spread model. The model incorporates the parameters of fuel load, live moisture content, terrain slope, and wind direction. The combined ignition probability from all burning neighbours is resolved through an independent probability formulation, capturing the cumulative influence of multi-directional fire fronts.

WRaP operates across two phases. In the first, a pre-fire preventive phase, a Monte Carlo ensemble of CA simulations generates two data layers: an ignition probability heatmap and a damage potential map identifying regions most at risk on fire start. In the second, active fire phase, WRaP transitions to real-time fire progression modelling. A computer vision (CV) pipeline operating on near-real-time (NRT) VIIRS satellite observations injects state corrections into the running CA at each satellite overpass, suppressing compounding drift errors that would otherwise cause purely model-driven simulations to diverge from ground truth.

The system is implemented as a Spring Boot 4.0 REST API with a layered Java architecture comprising twelve packages. Environmental inputs are sourced from ESA WorldCover 2021 for land cover classification, Sentinel-2 derived GeoTIFF bands for vegetation and moisture indices, ERA5 reanalysis wind fields, and OpenStreetMap road geometry for ignition likelihood estimation. Validation is conducted against the February 2023 Aberdare National Reserve wildfire event Hausdorff perimeter distance, and centroid displacement metrics. Preliminary results show a Hausdorff deviation of 3 cells and sub-cell centroid displacement, suggesting the model correctly captures the dominant spread behaviour of the event.

A successful outcome represents the first validated Rothermel-embedded CA fire spread model designed specifically for the East African ecosystem, providing a computationally accessible, physics-informed fire intelligence tool for forest management agencies such as the Kenya Forest Service.

Acknowledgement

The author wishes to acknowledge the supervisory guidance provided throughout this project. Appreciation is also due to the open-source communities behind GeoTools, Spring Boot, and the ESA Copernicus programme, whose freely available tools and datasets made this work feasible. The Kenya Forest Service and the broader wildfire management community in East Africa provided the real-world context that motivated this research.

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Abbreviations

Abbreviation	Full Form
CA	Cellular Automaton
CV	Computer Vision
DEM	Digital Elevation Model
ERA5	ECMWF Reanalysis v5
ESA	European Space Agency
FIRMS	Fire Information for Resource Management System
GeoTIFF	Georeferenced Tagged Image File Format
NDMI	Normalised Difference Moisture Index
NDVI	Normalised Difference Vegetation Index
NWP	Numerical Weather Prediction
NRT	Near-Real-Time
OSM	OpenStreetMap
ROI	Region of Interest
ROS	Rate of Spread
UTM	Universal Transverse Mercator
VIIRS	Visible Infrared Imaging Radiometer Suite
WRaP	Wildfire Risk and Progression Modelling

1. Introduction

1.1 Motivation

Forest fires have been, and remain as one of the major ecological agents in forests around the world. Using Kenya as a case study, the first 58 days of the year 2025 experienced 180 wildfire incidents, with over 3,350 acres of vegetation and significant water catchment areas around protected areas in Mount Kenya, and the protected Maasai Mau (Mercy M., et al, 2025) being significantly affected.

As a result, multiple approaches to the mitigation of uncontrolled forest wildfires have been implemented with varying degrees of success. This paper proposes the integration of Cellular Automata (CA) computational models for modelling fire rate of spread (ROS) and expected perimeter. By doing so, responsible bodies, such as the Kenya Forest Service (KFS) in Kenya will be enabled to both assess for areas with high ignition probability thus enabling more efficient allocation of early detection systems, and assess for fire risks in the case of a fire by monitoring its expected perimeter. That is precisely the offering of WRaP modelling; a duo-phased intelligence system that combines Computer Vision (CV) and physics informed CA for detection, prediction and prediction support.

The decision to opt for CA as the backbone for the fire perimeter modelling system is due to its intrinsic capabilities to model highly complex systems. Since its conception as a field of computing by Stanislaw Ulam and John Von Neumann, CA has been used to model increasingly complex systems across fields of physics, biology and computer science. This success has been largely due to its embarrassingly parallelizable, yet reliably accurate nature that makes it a very computationally viable option. The very nature of forest fires makes them a perfect candidate of CA modelling; this is largely because the nature wildfire ROS is dependent on very specific state transitions influenced by the state of neighbouring vegetation.

A successful implementation of such a solution would mean improved allocation of early wildfire detection systems already in place such as smoke detectors. Furthermore, it would allow responsible teams to assess risks that would be posed by a fire's progression, such as proceeding towards protected areas. Ultimately, this project aims to foster a substantial improvement in overall public safety and cultivate a far more coordinated and effective emergency response ecosystem, positively impacting Kenya's diverse geographic and infrastructural landscape.

Other aspects positively impacted by WRaP 's implementation shall be:

- **Community:** By providing accurate fire spread intelligence, WRaP shall enable timely evacuation warnings, protecting people and settlements bordering forest reserves. It safeguards the ecosystem services such as water provision and clean air that millions of Kenyans rely on.

- **Economic/Business:** Forest fires cause devastating economic losses, destroying timber assets, disrupting tourism, and straining public budgets. WRaP's focus on prevention and rapid containment helps safeguard these national assets, supporting sustainable economic development.
- **National Development:** The system aligns directly with Kenya's national vision for environmental conservation and climate action by ensuring the long-term health of critical natural infrastructure. It solves the authentic, real-world problem of resource scarcity by ensuring every KFS resource is used optimally where the risk is highest.

1.2. Statement of Problem

On February 6th, 2023, A forest wildfire suspected to have been started by illegal human activity was started. Just a week after its ignition, the wildfire had claimed 2 lives, one of a KFS responder and another of a fence attendant, and claimed over 40,000 acres of the Aberdare National Reserve. This case matches an emergent trend where drier seasons such as the February to March are becoming more severe and prolonged (Bancy M, et al, 2023), which in turn has resulted in far much more fierce and brutal wildfires that are significantly harder to manage.

These challenges are not localized to Kenya alone. In East Africa for instance, Ethiopia experienced a total of 10,773 wildfires in 2025 alone, which was a 12% increase from the previous year's 9,415 while Tanzania experienced 23,169 wildfires, which marked a 22.8% increase from the previous year's 19,646 (OurWorldinData, 2025). The ability for first responders to respond early and contain wildfires in time is significantly hampered by the economic conditions of these countries which limits their access to advanced early fire detection systems enough to cover their vast forested reserve regions. Case in point, Kenya's Mau Forest spans an estimated 995,834 acres, without intelligence to guide asset allocation on targeted hotspots, the early fire detection systems remain heavily underutilized, and their coverage suboptimal.

Compounding to this is the rather notable lack of fire modelling systems designed for the African tropical and Savannah ecosystems. While there exist advanced CA based fire modelling systems such as the FLARE wildfire systems designed for the Americas' ecosystems and the PROPAGATOR CA based wildfire simulator designed for the Mediterranean ecosystems (Andrea Truccia, et al.), these systems are designed to different fuel types, moisture regimes and climatic conditions. African grasslands and Afromontane forests that largely characterize protected zones in Kenya and neighbouring countries remain largely underrepresented in this front.

Kenya, and East Africa at large, remain largely reliant on agriculture as their economies' backbone, employing over 80% of the population (depending on the country) and accounting for 30 to 50% of the regions Gross Domestic Product (GDP) (Moses B, 2024). As such, since droughts are becoming fiercer and forest wildfires become more aggressive, proactive technological measures will have to be placed to secure the livelihoods and the food security of the region.

1.3 Proposed Solution

Overall Architecture

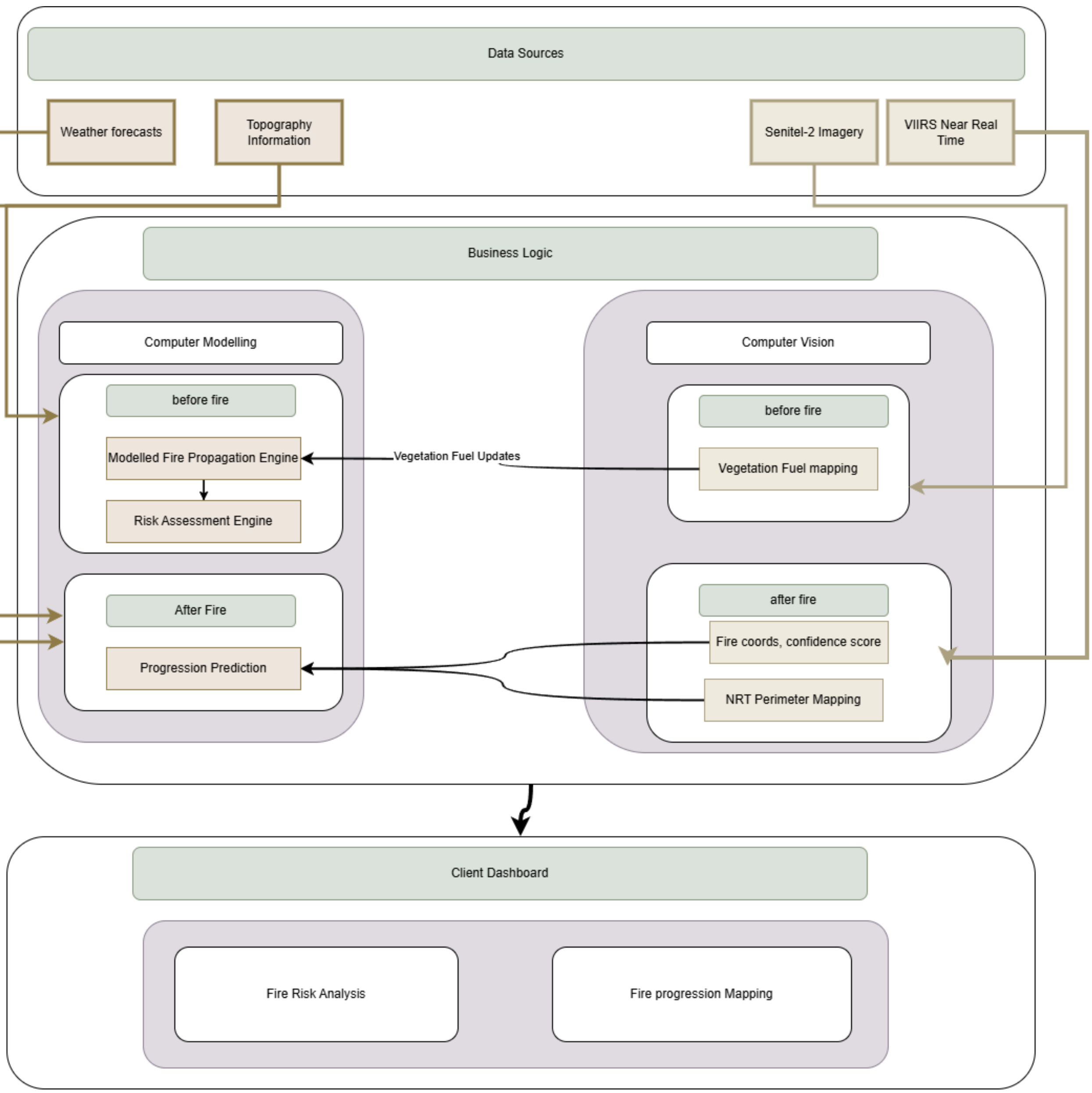

Figure 1: Overall architecture diagram for the WRaP Modelling software

The proposed solution is an integrated forest-fire intelligence system that brings together computer vision, predictive modelling, and real-time environmental data into one cohesive platform. The system is designed to support the entire fire-management lifecycle from early risk identification to active-fire monitoring and post-fire evaluation. To enable this, the system utilizes two phases:

Phase 1 - Before Fire (Prevention and Preparedness): WRaP allows responsible bodies to remain vigilant by assessing forest regions to identify areas with high ignition probability and high damage potential. This enables strategic deployment of fire detection sensors, preventive measures such as controlled burns and undergrowth removal, and informed resource allocation across vast forest reserves.

This phase provides an annotated heatmap to the responsible bodies to allow utilize reasonable measures.

Phase 2 - During Fire (Real-Time Response): The second phase is triggered when a fire is detected. The focus shifts from providing prevention support to providing insights on fire spread to support fire containment. WRaP models the expected fire progression over time. The system will continuously update its predictions using near real-time (NRT) satellite data, accounting for firefighting efforts and changing environmental conditions

At its foundation, the system relies on a continuous flow of satellite and environmental data including Sentinel-2 optical imagery, VIIRS near-real-time thermal observations, detailed topographic information and weather forecasts. These diverse datasets are fed into a central processing.

Within the core business logic, two major engines shall work together: the computer vision pipeline and the fire-modelling framework. The computer vision component will be focused on interpreting satellite imagery to understand the current state of the forest while the CA fire-modelling component uses the processed data to predict risk and simulate fire behavior. Prior to any fire event, the fire-modelling component combines vegetation conditions, weather forecasts, historical fire patterns, and topographical influences to estimate ignition probabilities and identify zones of high damage potential *(As seen in img2)*. This is achieved through a hybrid modelling approach that integrates Cellular Automata for rule-based fire spread. Once a fire is active, the model shifts to real-time progression prediction using continuously updated fire-
perimeter data and environmental variables to generate hour-by-hour forecasts of how the fire is likely to progress *(As seen in Img3)*. These predictions adjust automatically as new satellite observations become available, allowing the model to remain adaptive and reflective of changing conditions.

All outputs that are vegetation maps, fire perimeters, risk assessments and spread predictions are delivered through an intuitive dashboard designed for use by forest agencies, emergency responders, and environmental authorities. The dashboard presents the information visually through layered heatmaps, map-based overlays, and time-stamped simulations of predicted fire paths. By integrating all aspects of monitoring, detection and forecasting into a single system the solution provides a powerful, data-driven platform that supports proactive prevention, informed decision-making during active fires and long-term environmental management.

Computer Modelling Overview

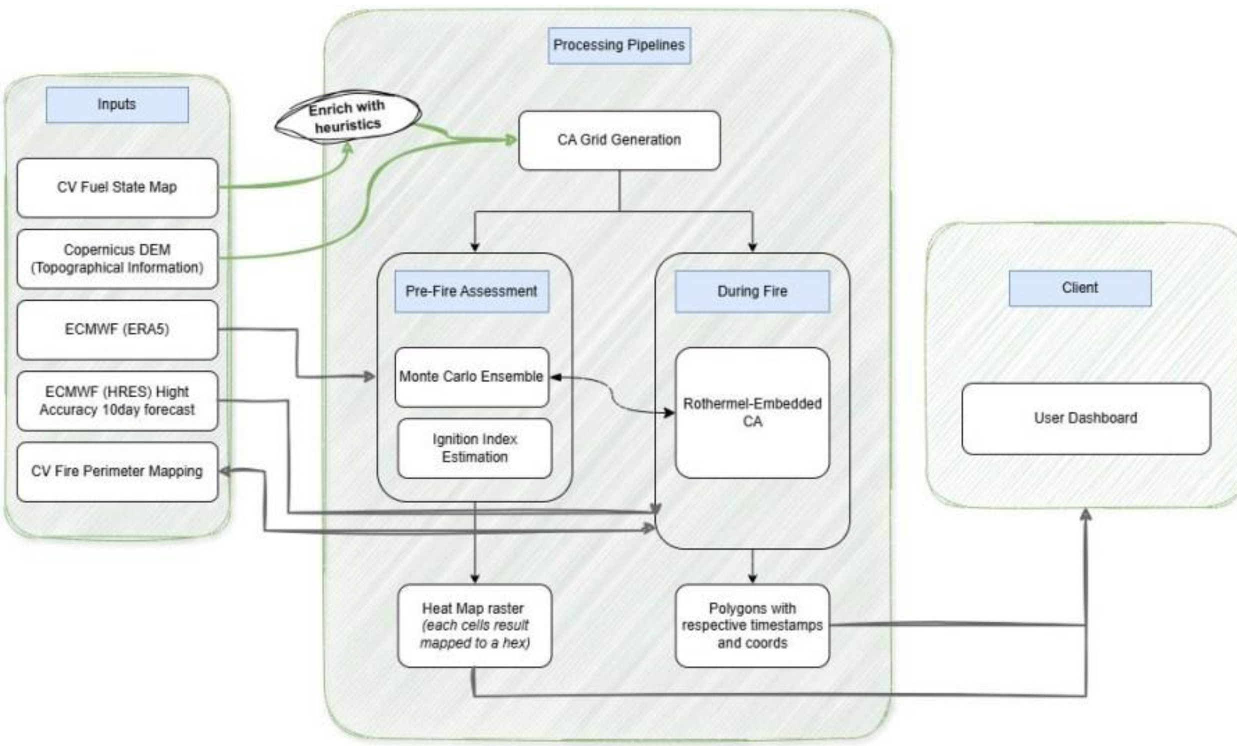

Figure 2: Computer Modelling operations overview diagram

The data layer pulls together all relevant geospatial inputs: vegetation and moisture imagery, elevation models, human-activity maps, and both historical and live fire detections. These
datasets flow in automatically through a scheduled Python ETL pipeline and sit in cloud storage for reliable access.

A Preprocessing Layer converts raw inputs into structured spatial products. This involves aligning resolutions, deriving vegetation condition metrics, and building analysis-ready grids. These cleaned and standardized layers supply both risk assessment and real-time modelling components.

The Risk Assessment module delivers two key outputs.

1. Ignition Risk Mapping identifies where fires are most likely to spark by weighing vegetation, moisture, weather, and human factors.
2. Damage Potential Mapping gauges how severe a fire could grow at each location, offering a high-level proxy for worst-case spread and resource-allocation priorities.

Together, these maps shape prevention strategies and drive preparedness planning.

The Real-Time Progression component forecasts how an active fire will unfold after ignition. It draws on the structured environmental grid, current fire perimeter, and incoming meteorological feeds to simulate forward spread. The model refreshes continuously as new satellite detections stream in, adjusting the system state so forecasts mirror actual ground conditions.

1.4. Objectives

1.4.1 main Objective

To design and develop a dual-phase, physics-informed wildfire risk and progression modelling system (WRaP) that integrates a Rothermel Rate of Spread embedded Cellular Automata engine with a satellite computer vision correction pipeline, providing pre-fire ignition risk intelligence and real-time active fire spread forecasting for East African forest ecosystems.

1.4.2 Primary Objectives

i. To design a computer vision system capable of continuously monitoring forests by mapping vegetation density, classifying vegetation states, detecting active fire perimeters, and analysing temporal vegetation changes from satellite imagery.

ii. To develop a predictive fire-risk assessment model that estimates ignition probability and potential damage using historical fire data, vegetation conditions, environmental factors, and topographical characteristics.

iii. To design and develop a real-time fire progression modelling framework that forecasts fire spread using cellular automata, integrating weather forecasts, topography and near real-time satellite updates.

iv. To evaluate the accuracy, responsiveness, and reliability of monitoring, risk assessment and fire progression systems through historical validation, real-time testing, and comparison against known fire events.

v. 5. To develop an intuitive user interface that visualizes vegetation states, risk maps, and fire progression predictions through interactive heatmaps and time-stamped overlays.

2. Literature review

2.1 Background

2.1.1 Origins and Fundamentals of Cellular Automata

Cellular Automata (CA) are dynamic computational models in which both space and time are discrete, consisting of an arrangement of cells on a regular lattice where each cell can occupy a finite number of states. The state of each cell at a given time step is determined by a transition rule applied to the cell's own current state and the states of its neighbouring cells (Von Neumann, 1966). It is precisely this local-rule, global-emergence property that makes CA a uniquely powerful tool for simulating complex spatial systems.

The concept was jointly developed in the late 1940s by Stanislaw Ulam and John von Neumann at Los Alamos National Laboratory. Following Ulam's suggestion to adopt a discrete, cell-based formalism, von Neumann formalised the cellular automaton in his landmark posthumously published work, Theory of Self-Reproducing Automata (Von Neumann, 1966). His 29-state two-dimensional automaton demonstrated that local interaction rules alone are sufficient to produce complex, self-reproducing global behaviour; a result that would prove foundational to computational modelling across disciplines.

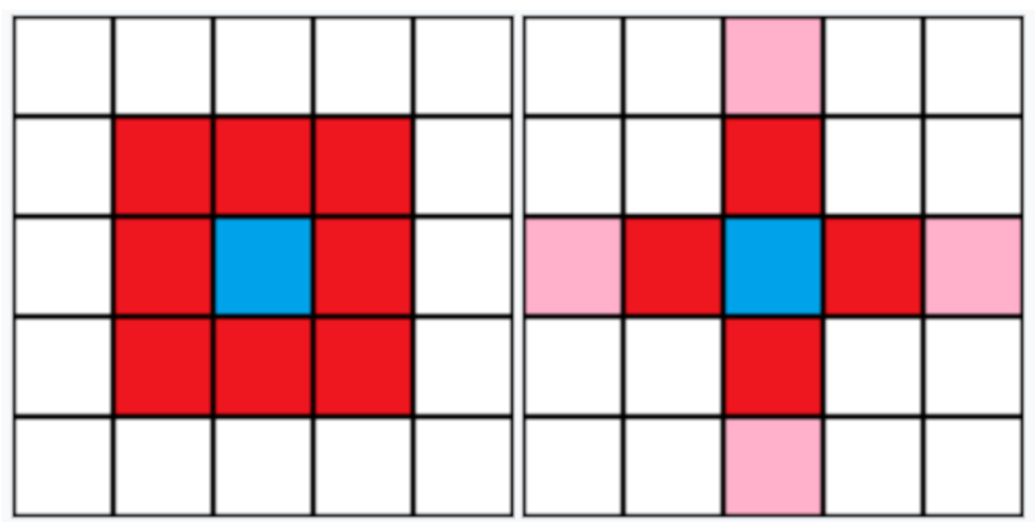

Figure 4 (left): Demonstrates the Moore neighbourhood distribution for the blue cell which identifies eight neighbours | (right): Demonstrates the Von Neumann neighbourhood distribution for the blue cell where in both images the red and pink box

The two most widely used neighbourhood configurations in two-dimensional CA are the von Neumann neighbourhood, in which each cell interacts with its four orthogonal neighbours *(See img5a)*, and the Moore neighbourhood, in which each cell interacts with all eight adjacent cells including diagonals *(See img5b)*. For wildfire modelling, the Moore neighbourhood is typically preferred because it allows fire to propagate in eight directions on a square grid, more closely approximating the continuous, radially expanding nature of a real fire front (Alexandridis et al., 2008).

2.1.2 Cellular Automata in Physical and Environmental Modelling

The applicability of CA to real-world physical and environmental phenomena is rooted in their embarrassingly parallelisable structure and their capacity to represent spatially distributed, state-transition-driven processes. The very characteristics of wildfire spread; where each parcel of land transitions through discrete states (unburned, burning, burned) based on the burning state of its neighbours, local fuel conditions, wind, and topography. This is demonstrated here *(see img6 blow)*, cells that cannot be burned i.e. Water bodies can be represented in a blue state.

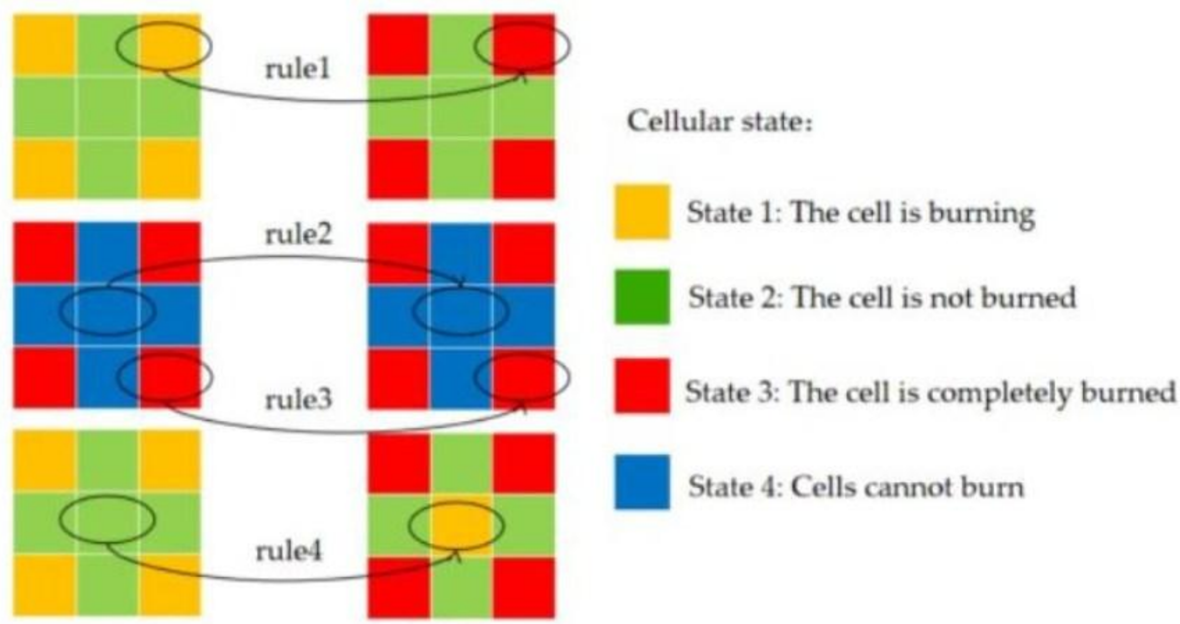

Figure 5 Cell state transitions in a CA fire spread model

CA models fill a significant gap in the landscape of wildfire simulators. Unlike purely deterministic physical models, which require intensive preliminary studies to parametrise complex fluid-dynamic and combustion equations, CA models operate on relatively simple local transition rules that can nonetheless produce emergent global fire behaviour consistent with observation. Unlike purely empirical models, which tend to be static and ecosystem-specific, CA frameworks are modular and extensible: environmental variables such as wind speed and direction, terrain slope, vegetation type, and fuel moisture content can be incorporated as probabilistic modifiers of the cell state transition rules, allowing the model to adapt to different spatial contexts (Sullivan, 2009).

2.1.3 Foundational Fire Physics: The Rothermel Rate of Spread Model

The physical basis for wildfire rate of spread (ROS) modelling is widely rooted in the quasi-empirical model developed by Richard C. Rothermel at the Missoula Fire Sciences Laboratory. Published in 1972 in his paper "A Mathematical Model for Predicting Fire Spread in Wildland Fuels," Rothermel's model computes the steady-state surface fire spread rate and intensity as a function of fuel physical and chemical properties (loading, depth, particle surface-area-to-volume ratio, heat content, mineral content, and moisture content) and environmental inputs (mean wind velocity and terrain slope) (Rothermel, 1972). Fifty years after its publication, the Rothermel model remains the most widely used tool for wildfire behaviour prediction worldwide, embedded in dozens of operational fire management systems including BEHAVE, FARSITE, and the US National Fire Danger Rating System (Andrews, 2018).

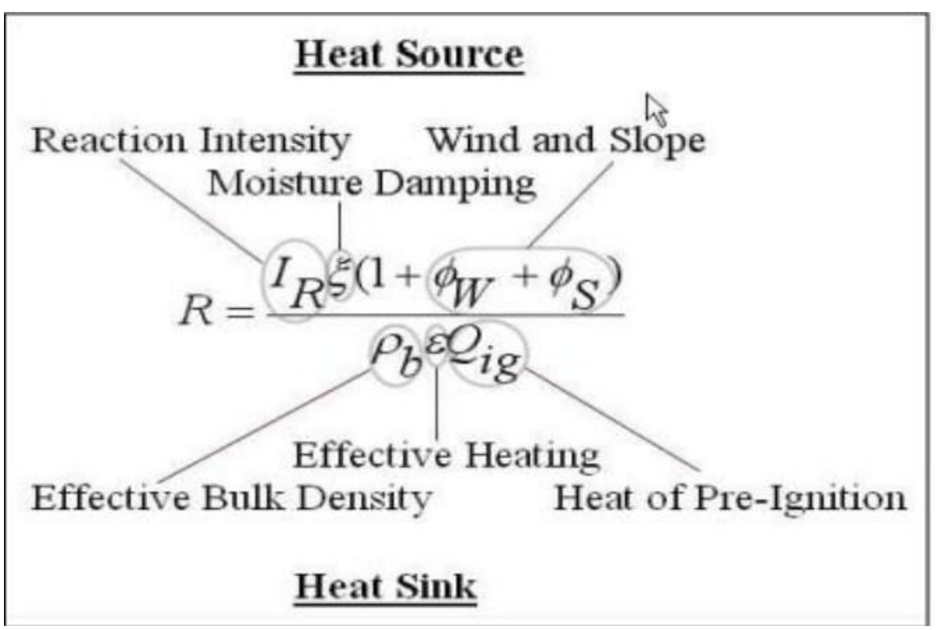

Figure 6A breakdown of the Rothermel's ROS model. (Robert Ziel, et al. 2009)

2.1.4 Cellular Automata-Based Wildfire Spread Modelling

The PROPAGATOR (Trucchia et al., 2020) represents one of the most operationally mature CA-based wildfire simulators. It models stochastic fire propagation through probabilistic transition rules modified by topography, wind, and fuel moisture, runs independent simulation ensembles to generate cell-level burn probability maps, and integrates Numerical Weather Prediction data for operational deployment in Mediterranean regions

Cell2Fire (Pais et al., 2021) represents a parallel development, building a CA-based fire growth simulator on the Canadian Forest Fire Behaviour Prediction (FBP) system and targeting operational use in Canada, Spain, and Chile. A 2025 study extended Cell2Fire to US landscapes and demonstrated high agreement with established simulators such as BEHAVE and FARSITE across both synthetic and heterogeneous real-world landscapes, while applying BlackBox Optimisation (BBO) to fine-tune ROS adjustment factors against real burn scars (Vargas et al., 2025). The critical observation from this body of work is that CA simulators achieve far greater computational efficiency than vector-based perimeter expansion models, which suffer from fire crossovers and unburned islands and scale poorly for large, prolonged fires, while remaining accurate enough for operational use when appropriately calibrated.

2.1.5 Wildfire Risk and Ignition Probability Assessment

Wildfire risk assessment frameworks generally seek to quantify two complementary dimensions: the probability that a fire will ignite at a given location (ignition risk), and the likely severity or spatial extent of a fire should it start (damage potential). WRaP's dual-phase architecture maps directly onto this distinction, with Phase 1 combining both dimensions into a pre-fire risk intelligence layer.

The primary input variables driving ignition risk in the literature include vegetation type and density, fuel moisture content, meteorological conditions (temperature, wind speed, relative humidity), topographic slope and aspect, and human activity proximity (Jain et al., 2020). Remote sensing provides scalable access to most of these variables at the landscape scale. A comprehensive review of wildfire risk modelling approaches (Jain et al., 2024) found that while traditional danger-rating systems such as the Canadian FWI provide foundational indices, machine learning approaches; particularly ensemble methods and deep learning, consistently outperform them in capturing the non-linear interactions between predictors. For WRaP's risk assessment module, this supports the use of a data-driven approach that combines satellite-derived vegetation and moisture indices with historical fire occurrence data and meteorological inputs.

2.2 State of the Art

The area of research of wildfire modelling has experienced rapid transformation with advances in machine learning, data assimilation and high-performance computing. These converging trends have been

2.2.1 Deep Learning and Neural Approaches to Fire Spread

One of the most significant emerging trends is the integration of deep learning into fire spread prediction. Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and graph neural networks have all been explored as either replacements for or complements to classical CA and physics-based models. An example of this is the use of deep learning models trained on historical fire progression imagery to predict next-step fire perimeters directly from satellite inputs, bypassing the need to explicitly model underlying physical spread mechanisms (Jain et al., 2020)

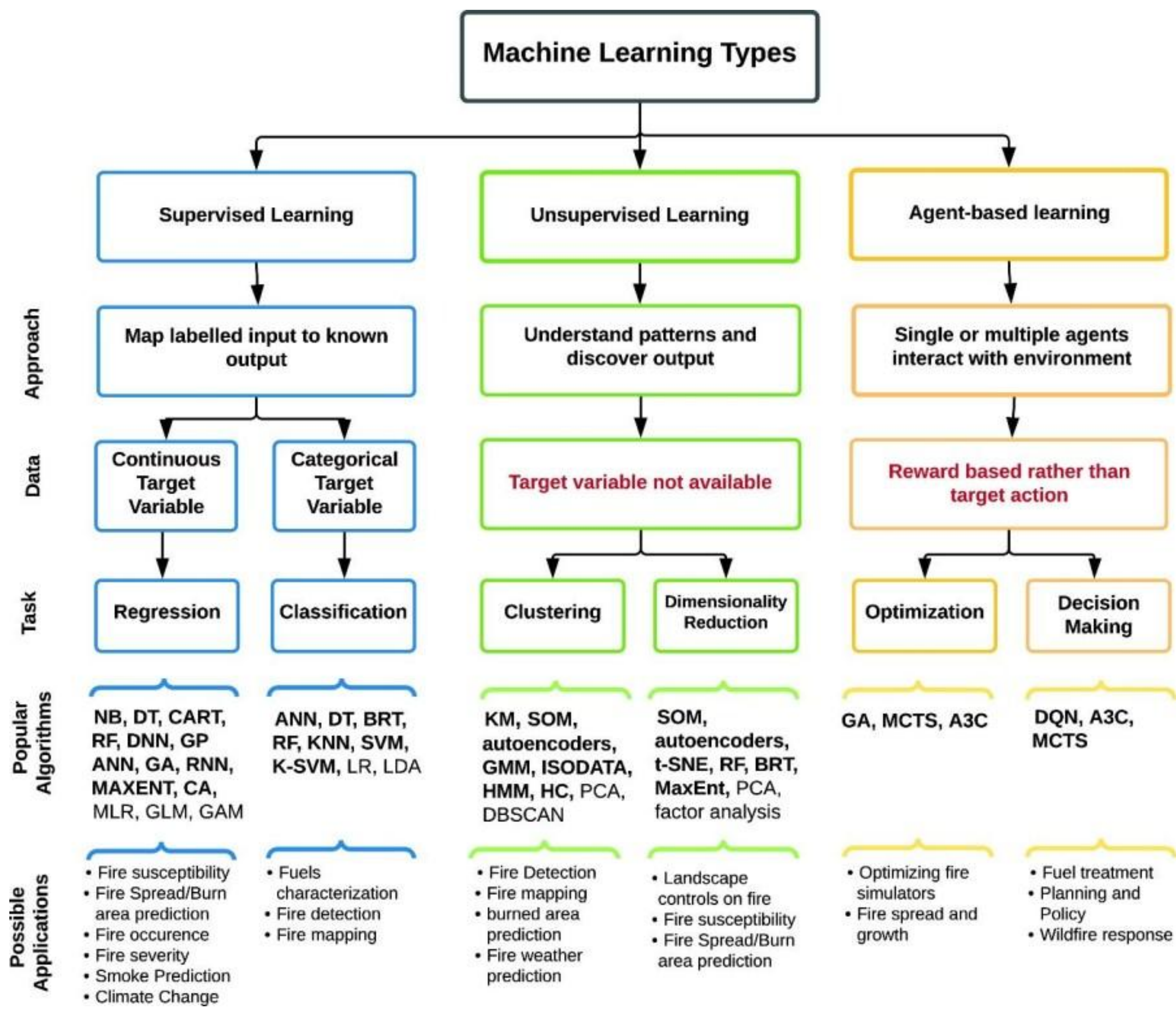

Figure 7Machine learning approaches and applications in wildfire science

While these approaches achieve strong predictive performance in data-rich contexts, they inherit the data-dependency limitation that characterises the broader class of historically-trained models (Iqbal H Sarker, 2021). A consequence of this is models trained with data regions with less rigorously documented wildfire events are more prone to overfitting, a phenomenon where the model learns noise instead of capturing relationships.

2.2.2 Data Assimilation and Observation Driven Model Correction

Data assimilation, the process of systematically incorporating observational data into a running model to correct its state, is an established technique in atmospheric and hydrological forecasting that is now being introduced into the fire modelling space. There has been work exploring ensemble-based assimilation methods, including variants of the Ensemble Kalman Filter (EnKF), applied to wildfire spread models as a means of reconciling simulated fire fronts with satellite-detected perimeter observations during active fires (Rochoux et al., 2014). This body of research establishes a formal computational framework for exactly the observational correction mechanism that WRaP implements through its CV-CA coupling: the computer vision module produces a perimeter observation at each satellite overpass, and this observation is used to update the CA's internal state, analogous to an assimilation step. Framing WRaP's architecture within this data assimilation paradigm connects the system to a mature body of theoretical and applied literature and provides a pathway for formal uncertainty quantification in future work.

2.2.3 3-D Fire Modelling to Capture Spotting Probability Due to Fire bands

While 2-dimensional CA and semi-empirical models offer computational efficiency, they fall short when it comes to accounting for vertical transport mechanisms that drive more complex wildfire behaviours such as spotting. State of the art systems such as the QUIC-FIRE (Linn et al., 2020; 2023), take a 3-dimensional approach that better simulate the behaviours. In as much as they may be computationally expensive, they significantly improve from the older approaches such as the FIRETEC model by replacing full Navier-Stokes solvers with a 3D diagnostic wind solver (QUIC-URB) coupled to a physics-based Cellular Automata (Fire-CA). This allows for 3D interactions such as the influence of heterogenous fuel structures on wind -fields simulated in real time

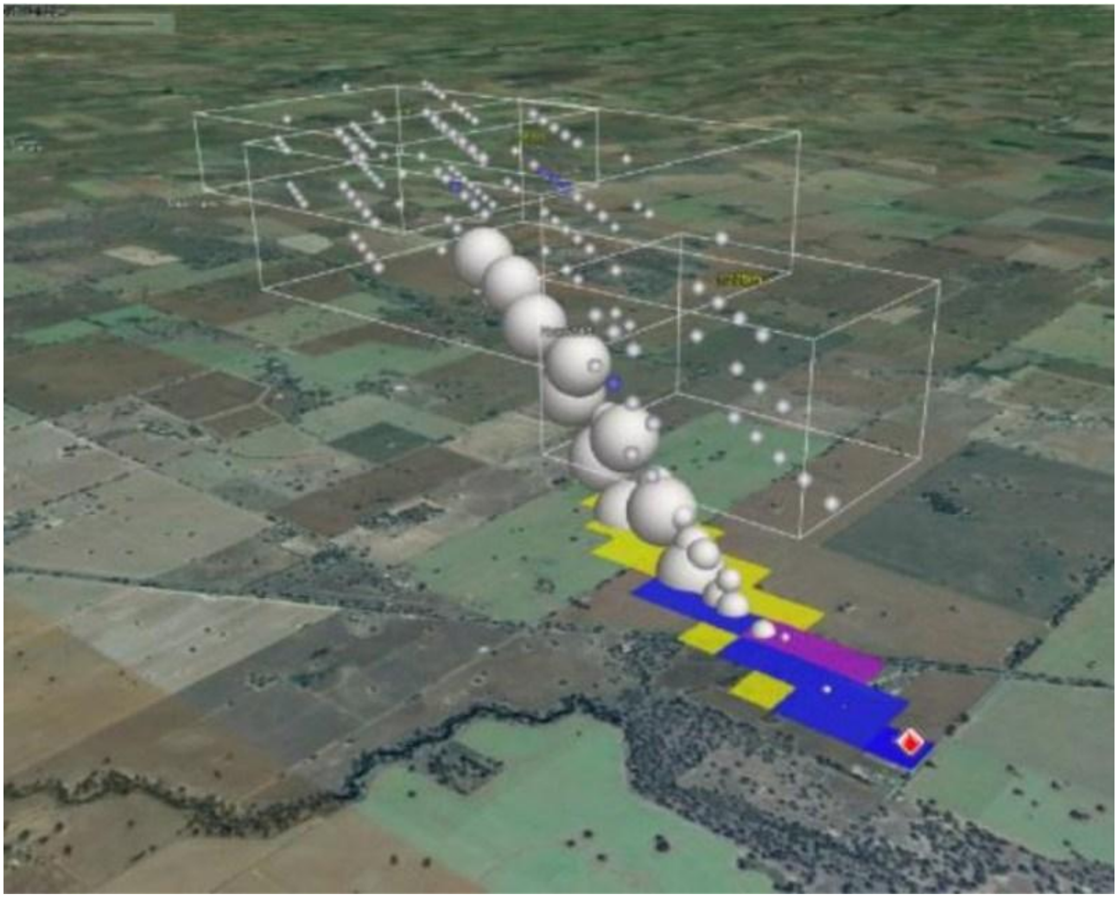

Figure 8QUIC-FIRE 3D fire band simulation

2.3 Summary (Gap Analysis)

While existing research has made significant progress in individual components (CV-based fire detection, CA modelling, risk optimization), several gaps remain:

2.3.1 Historical Data Dependency

Emergent approaches to higher accuracy fire modelling CA by implementing LSTM models to learn non-linear state transition rules more accurately inherently require a larger volume of dataset. This is a significant bias towards less high-quality and extensive historical datasets covering fuel types, moisture regimes and fire spread as it is in the Western United States, or Mediterranean Europe. However, regions such as East Africa where such comprehensive historical data is often sparse or absent remain under represented. WRaP addresses this by a CA-CV couple that iteratively correct itself with a goal to suppress n compounded errors.

2.3.2 Unaccounted Real-World State Changes During Wildfire

Once a fire is actively burning, numerous factors alter the state of the landscape in ways that diverge from what a purely model-driven simulation can anticipate. These include fire fighting efforts such as including water drops, firebreak construction, and controlled back-burns, and natural factors such a fire spotting due to fire bands.

Purely theoretical and mathematical models such as the PROPAGATOR, Cell2Fire, and FARSITE do not possess a mechanism for detecting these ground-truth deviations and correcting their internal state accordingly; they continue propagating from their last known state, which progressively diverges from reality (Sullivan, 2009; Trucchia et al., 2020). State of the art approaches such as the Australia's FLARE Wild Fire research approach the same by taking a 3-dimensional approach that accounts for changes to fire behaviour due to natural conditions such as spotting by modelling the fire band path.

WRaP resolves this through its computer vision module, which maps the active fire perimeter from VIIRS thermal detections at each satellite overpass and feeds this observed perimeter back into the CA engine as an authoritative ground-truth correction. This closes the feedback loop between simulation and observation, ensuring that the spread model continuously reflects the actual state of the fire rather than an increasingly stale forward projection.

2.3.3 Dual Risk Assessment

Existing systems typically output a single composite fire danger or risk index that conflates ignition likelihood with potential fire severity. This limits their actionability: an area may carry high ignition probability due to human activity but low damage potential due to sparse fuel, or conversely high damage potential due to dense dry vegetation but low ignition probability due to remoteness. WRaP separates these dimensions explicitly, providing independent ignition probability and damage potential maps that allow agencies to make more targeted resource allocation decisions.

3. Approach and Methodology

3.1 Description

The project followed a sequential implementation order structured around package dependency. Each group of classes was implemented and tested before the next group was started, ensuring that higher-level packages always consumed stable, contracted interfaces. The seven implementation stages were:

1. Literature review and fuel model parameter derivation for East African vegetation types.
2. Geospatial data pipeline design: ESA WorldCover, Sentinel-2 CV GeoTIFF, ERA5 wind, OSM roads.
3. CA grid domain design: 100 m cell resolution, Moore neighbourhood, four discrete states.
4. Phase 1 engine: ignition likelihood index construction, Monte Carlo ensemble, risk map assembly.
5. Phase 2 engine: Rothermel ROS integration, active frontier tracking, CV correction injection.
6. REST API layer: Spring Boot endpoints, JSON response design, CORS for frontend.
7. Validation: Rothermel reference value checks, Phase 1 risk map plausibility assessment.

3.1.1 Grid Design

The landscape of interest is discretised into a two-dimensional grid of square cells. Each cell is assigned one of four states: UNBURNED, BURNING, BURNED, or NON-COMBUSTIBLE. Cell size is set to 100 metres, representing a practical balance between Sentinel-2's 10–20 metre native resolution and the computational cost of simulating large forest*)*. Each cell is also assigned a static environmental state vector derived from the CV input layer, comprising:

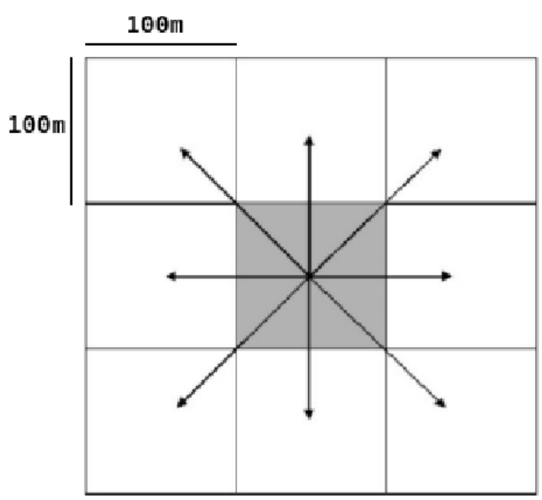

Figure 9: 100m CA grid cell with eight Moore spread directions

(1) *fuel load, proxied by the Normalised Difference Vegetation Index (NDVI*; a remote sensing metrics ranging from -1 to 1 used to quantify green vegetation. A rough interpretation of its values would be 0.6 to 0.8: High density, healthy vegetation, 0.2 to 0.3: Shrub and grasslands, 0 to 0.1: Bare soil, rocks, or deserts, and Water, snow, or clouds);

(2) *live fuel moisture content (LFMC,* a ratio of water to dry matter in living vegetation ranging from 30% to 300% that indicates to ignition potential), proxied by the Normalised Difference Moisture Index (NDMI);

(3) terrain slope magnitude and aspect, derived from the Copernicus Digital Elevation Model; and

(4) vegetation type classification. Non-vegetated cells i.e., water bodies, bare rock, built areas — are initialised as NON-COMBUSTIBLE and excluded from all transition calculations.

3.1.2 Phase 1: Pre-Fire Risk Assessment (Monte Carlo Burn Probability Ensemble)

Prior to any active fire event, WRaP generates a risk intelligence surface covering the area of interest. This is achieved through a Monte Carlo burn probability ensemble, which reuses the Phase 2 CA engine directly rather than requiring a separate analytical model.

A seasonal wind climatology field is first derived for the target area from ERA5 reanalysis data, providing representative mean wind speed and direction per grid cell for the dry season period. This field is held fixed across all ensemble runs, as it represents the climatological fire-weather conditions under which the risk assessment is valid.

An ignition likelihood index I(c) is then computed for each cell c as a simple weighted function of three observable variables: (1) vegetation dryness, measured by normalised NDMI; (2) historical fire occurrence density, derived from the NASA FIRMS archive of VIIRS and MODIS detections over the preceding five years; and (3) proximity to human activity, derived from OpenStreetMap land-use layers. This index is used solely to define the spatial sampling distribution for ignition seeding — it is not itself a risk output.

N ignition seeds are then sampled from the grid with probability proportional to I(c). For each seed, the Phase 2 CA engine (described in Section 3.1.3) is run forward for a fixed simulation horizon of 24 hours using the seasonal wind climatology field and the static environmental state vectors. After all N runs are completed, two outputs are derived: (1) the Damage Potential Map, in which each cell's value is its burn frequency across the N runs. The fraction of simulations in which it was reached by fire; and (2) the Ignition Risk Map, which is the spatially smoothed ignition likelihood index I(c). Together these form the pre-fire dual-layer heatmap delivered to responsible bodies in Phase 1.

3.1.3 Phase 2: Active Fire Spread (Rothermel-Embedded CA)

Upon detection of an active fire event, signalled by the CV module from VIIRS thermal anomalies, the CA transitions to real-time spread simulation. The detected fire perimeter is used to initialise the BURNING cell set. The simulation then advances in discrete time steps of Δt minutes.

At each time generation t, all the cells check their Moore's neighbours for any cell with a BURNING state. If for a cell C_i one or more neighbours (N) are established to be in BURNING state, the burning neighbours are iterated over, and the following steps are applied:

First, the unit direction vector d_e from cell C_i to cell N_j is computed. For orthogonal neighbours this is one of the four cardinal directions; for diagonal neighbours it is one of the four 45° intercardinal directions, with the cell separation distance scaled by $\sqrt{2}$ accordingly.

Second, the effective wind component U_e (Computed as WindSpeed W x direction d_e) along direction d_e is computed by projecting the current wind vector onto d_e. Negative projections, indicating wind blowing against the spread direction, are clamped to zero as wind slow down is captured implicitly by the reduced ROS in upwind directions.

Third, the slope component φ_e is computed relative to C_i . This is computed by first getting the change in elevation of the neighbour relative to the cell $\boldsymbol{\Delta Z}$ **=** Elevation (N_j) – Elevation (C_i). Using distance D_j as CellSize x sqrt (2). The Slope S_j is computed as $\mathbf{tan^{-1} \times (\Delta Z / D_j)}$

Fourth, the Rate of Spread ROS_e (metres per generation step time i.e. 1 minute) toward cell C_i is computed using a simplified form of the Rothermel (1972) surface fire spread model, taking as inputs: the fuel load, moisture content, and vegetation type of cell j; the effective wind component U_e and the effective slope φ_e.

Fifth, the directional transition probability P_e is computed by first computing the ratio representing the total distance covered by:
(ROS x step time)/Distance: Where $P_e < 1$ represents the ratio of distance covered to not covered (this in our case represents probability of ignition, with higher values representing grater fire proximity, and a correspondingly higher probability of ignition. While $P_e > 1$ represents a rate of spread greater than the Distance which can be interpreted as 1). And the Distance is CellSize x sqrt (2)

The probability of ignition of the cell will be a consideration of probabilities of ignition from every N_j. Since all probabilities for ignition due to neighbours are independent, the probability that at least one ignites C_i is:

$$\mathbf{P(Ignition) = (1-(1-P_{e1})\,(1-P_{e2})\,\ldots(1-P_{ej}))}$$

The probability of ignition is collapsed to a state transition Boolean by the following logic:

A random number *rand* in the range 0 and 1 (exclusive) is generated.

The state transition shall be resolved to: *rand* < P(Ignition)

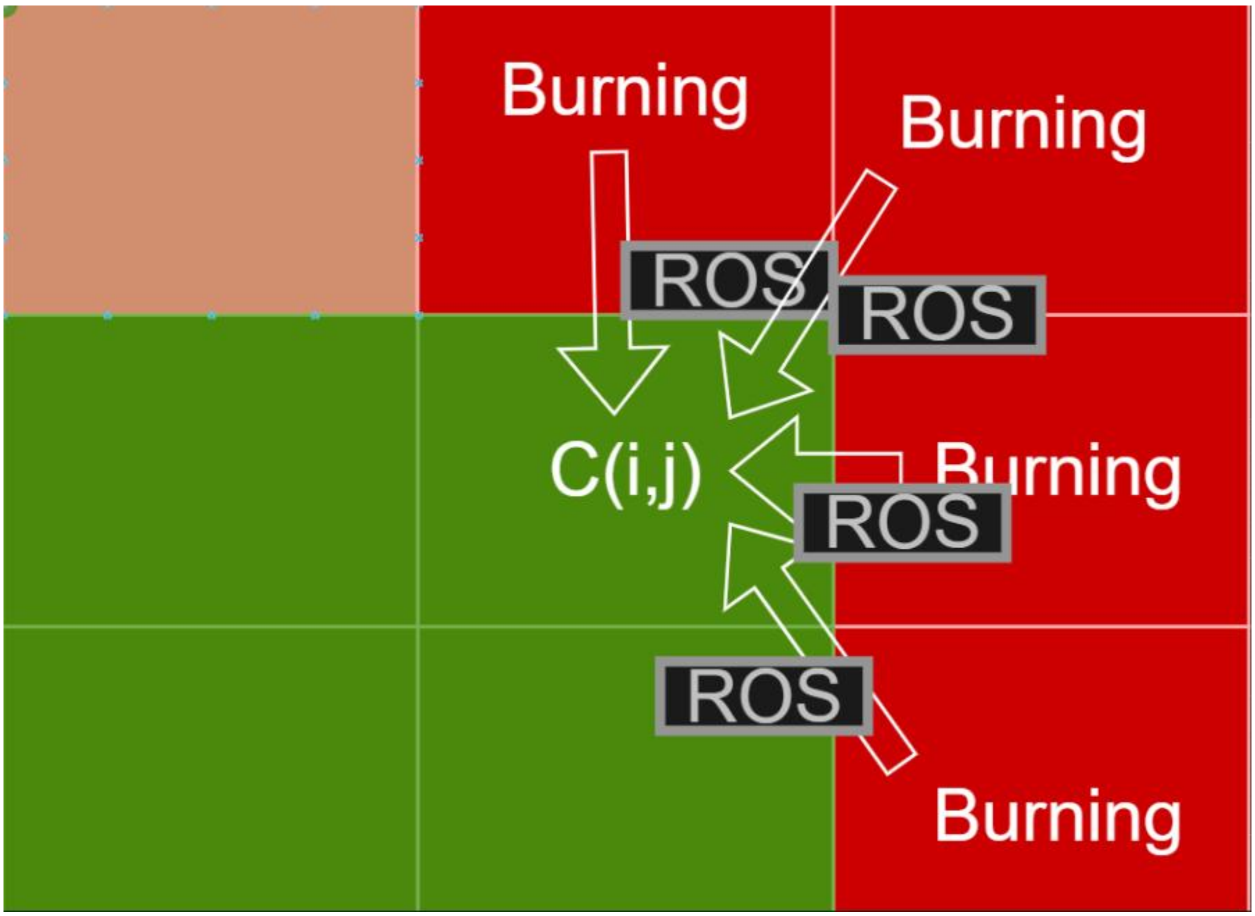

Figure 10. Demonstration of the ROS and probability of ignition being calculated from the BURNING states

3.1.4 CV Re-injection and State Correction

At each satellite overpass interval, the CV module delivers an updated observation layer to the CA engine. This layer contains three types of corrections: (1) cells confirmed as BURNED by thermal detection are forced to the BURNED state regardless of simulation state, overriding any in-progress burn; (2) cells where firefighting interventions have been detected — identified by the CV module from suppression signatures in the imagery are reset to NON-COMBUSTIBLE for the duration of the suppression window; and (3) fuel moisture values are refreshed for all UNBURNED cells from the latest Sentinel-2 observation. Following this correction step, the CA resumes propagation from the updated state. This closed feedback loop between simulation and observation is the mechanism by which WRaP prevents initialisation and drift error from compounding over the course of a multi-day fire event.

3.2 Technologies

Component	Technology / Tool
Language	Java 21 (Temurin 21.0.10 JDK)
Framework	Spring Boot 4.0.3 (Maven build)
Geospatial	GeoTools 31.0 (gt-coverage, gt-geotiff, gt-referencing, gt-epsg-hsql)
Math / Statistics	Apache Commons Math 3.6.1 (weighted sampling for Monte Carlo seeding)
Data Binding	Jackson + JavaTimeModule
Satellite Data	Sentinel-2 (via CV module GeoTIFF), VIIRS (thermal correction), ESA WorldCover 10 m
Wind Data	ERA5 reanalysis wind fields (scalar stub for development)
Road Geometry	OpenStreetMap pre-downloaded GeoJSON (ODbL licensed)
IDE / Build	IntelliJ IDEA, Maven 3.x, Java 21 compiler plugin
OS	Windows 11 Home (64-bit, x64)

Table 1: Technologies and tools used in WRaP

3.3 Data

3.3.1 CV Fuel State GeoTIFF

An 11-band Sentinel-2 derived raster delivered by the CV module. Bands consumed by the ingestion layer are NDVI (band index 5), NDMI (6), elevation (8), slope (9), and aspect (10). CRS is EPSG:32737 (UTM Zone 37S). Native pixel size is 10 m. The file is resampled to 100 m for the CA grid using block-averaging across all five continuous bands.

3.3.2 ESA WorldCover 2021

A 10 m globally consistent land cover classification product produced by the European Space Agency. Class codes are mapped to the engine's seven VegetationType constants via the EsaBandLayout constants class. Resampled to 100 m using majority-class algorithm with a tie-breaking rule that prefers combustible codes over non-combustible codes to preserve fire risk conservatism.

3.3.3 ERA5 Wind Fields

ECMWF ERA5 reanalysis 10 m wind data. For development, a scalar stub file provides uniform wind speed and FROM-direction across the entire grid. Format: JSON with speedMs and directionDeg fields. Wind direction follows meteorological convention (FROM-direction, clockwise from north). The midflame conversion factor (0.4) is applied once inside RothermelRosCalculator and nowhere else.

3.3.4 OSM Road geometry

GeoJSON of road and path linestrings extracted from OpenStreetMap for the deployment area. Highway tags retained: track, path, unclassified, tertiary. Reprojected to UTM 37S before storage. Used exclusively for road proximity computation in the ignition likelihood index. Built-up masking is handled by ESA code 50.

3.3.5 CV Fuel Risk Map

A raster with 3 values, low medium and high. This is consumed to provide analysis of higher risk areas in the region of interest

3.4 Ethical Considerations: P.A.U.S.E Framework

The PAUSE framework (Privacy, Accountability, Understanding & Explainability, Social impact, and Ethics of care) is applied below to identify and address the ethical dimensions of WRaP's design and deployment.

3.4.1 Privacy

The CA engine operates exclusively on geospatial environmental data (satellite imagery, elevation, weather) and does not process any personally identifiable information. The human activity proximity layer used in the ignition likelihood index is derived from aggregated OpenStreetMap infrastructure classifications, not from individual movement or behavioural data.

3.4.2 Accessibility and Inclusivity

WRaP is designed to be accessible to resource-constrained institutions in Kenya and East Africa. The system relies exclusively on freely available satellite data (Sentinel-2, VIIRS, ERA5, Copernicus DEM) and open-source software libraries, eliminating licensing costs that would otherwise be prohibitive for agencies such as the Kenya Forest Service. The CA engine is computationally lightweight enough to run on modest server hardware, and the outputs shall be provided in an intuitive and easy to use and interpret user interface. This design ensures that the system remains accessible to institutions with limited technical infrastructure.

3.4.3 User-Support

Deployment of WRaP will be accompanied by comprehensive documentation for responsible personnel and disaster management coordinators. Training materials will include: (1) a user guide explaining how to interpret risk heatmaps and perimeter forecasts; (2) worked examples demonstrating how to use the outputs to inform resource allocation and evacuation planning decisions; and (3) a technical annex describing the CA methodology, the Rothermel ROS model, and the assumptions underlying the predictions, written for an audience familiar with fire management but not necessarily computational modelling. An issue-tracking system will be established to allow users to report discrepancies between predicted and observed fire behaviour, enabling continuous model refinement.

3.4.4 Societal Equity

WRaP is designed to reduce wildfire harm to communities bordering Kenya's forest reserves, whose livelihoods and safety are disproportionately affected by uncontrolled fires. Its approach makes it computationally feasible even to more resource restricted countries and communities, allowing the to also benefit. Remote communities are often more severely affected by wildfires; the system's accessibility allows it to run on their computers however resource restricted to offer timely assistance and guidance.

3.4.5 Ethical Decision Making and Transparency

WRaP uses a physics-informed analytical model (Rothermel ROS) rather than a black-box machine learning model. Every output *(ignition probability, damage potential, and fire perimeter)* is traceable to its physical inputs and model equations. The deviation discourse document maintained throughout development records every deviation from the original proposal, ensuring transparency in design decisions. Biases inherent in the fuel model parameters, which are not field-calibrated for East African ecosystems, are explicitly acknowledged in documentation and flagged with recalibration guidance.

4. System Analysis and Design

4.1 System Analysis

Java 21 provides the language baseline. GeoTools 31.0 supplies raster I/O (gt-coverage, gt-geotiff), coordinate reference system handling (gt-referencing, gt-epsg-hsql), and GeoJSON geometry utilities. Apache Commons Math 3.6.1 provides the EnumeratedIntegerDistribution used for weighted Monte Carlo seeding. The Rothermel (1972) model is fully specified in published literature and can be implemented without proprietary data.

All geospatial input data are freely available under open licences: Sentinel-2 imagery is free-of-charge through ESA's Copernicus programme, ESA WorldCover 2021 is published under CC-BY 4.0, ERA5 wind reanalysis is available under the Copernicus Licence, and OpenStreetMap road geometry is available under the Open Database Licence. No software licence fees apply: Spring Boot, GeoTools, Commons Math, Jackson, and Lombok are all open-source.

4.1.1 Feasibility

Technical Feasibility

Java 21 provides the language baseline. GeoTools 31.0 supplies raster I/O (gt-coverage, gt-geotiff), coordinate reference system handling (gt-referencing, gt-epsg-hsql), and GeoJSON geometry utilities. Apache Commons Math 3.6.1 provides the EnumeratedIntegerDistribution used for weighted Monte Carlo seeding. The Rothermel (1972) model is fully specified in published literature and can be implemented without proprietary data.

All geospatial input data are freely available under open licences: Sentinel-2 imagery is free-of-charge through ESA's Copernicus programme, ESA WorldCover 2021 is published under CC-BY 4.0, ERA5 wind reanalysis is available under the Copernicus Licence, and OpenStreetMap road geometry is available under the Open Database Licence. No software licence fees apply: Spring Boot, GeoTools, Commons Math, Jackson, and Lombok are all open-source.

Operational Feasibility

The system is designed for offline deployment. The four-step CV cache fallback chain in CvApiClient means the server can start and run a full Phase 1 risk assessment using previously downloaded satellite data even when the CV module is unreachable. Missing road data returns an empty RoadLayer, zeroing out the road proximity term in I(c) without error. Missing ERA5 wind returns calm conditions (0 m/s, 0°) with a warning log. No internet connection is required at runtime after initial data download.

4.1.2 Technology Stack

Component	Technology	Version	Licence
Language	Java (Temurin JDK)	21.0.6	GPLv2+CE
Framework	Spring Boot	4.0.3	Apache 2.0
Geospatial	GeoTools	31.0	LGPL 2.1
Math / Statistics	Apache Commons Math	3.6.1	Apache 2.0
JSON Binding	Jackson + JavaTimeModule	Spring BOM	Apache 2.0
Boilerplate	Lombok	Spring BOM	MIT
Build	Maven	3.x	Apache 2.0
Land Cover	ESA WorldCover 2021	10 m	CC-BY 4.0
Fuel State	Sentinel-2 (via CV GeoTIFF)	10 m	Copernicus
Wind Data	ERA5 Reanalysis	—	Copernicus
Road Geometry	OpenStreetMap GeoJSON	—	ODbL
OS (dev)	Windows 11 Home	25H2 x64	Proprietary

Table 2: WRaP CA Technology Stack with version and licence information

4.1.3 System Requirements

Non-Functional Requirements

ID	Requirement	Priority
NF R1	Monte Carlo ensemble (N=200) must complete in a wall-clock time that allows a forest officer to receive updated risk maps within a practical session (~5 minutes on a 4-core laptop).	High
NF R2	SuppressedZoneRegistry.isActive() must be thread-safe for concurrent reads from the Monte Carlo ForkJoinPool without synchronisation on the hot path.	High
NF R3	API responses must not contain raw grid arrays or GeoTIFF bytes. All output is compact numeric flat arrays or GeoJSON strings.	High
NF R4	The system must degrade gracefully on missing data: CV unavailable → local cache; no roads → empty RoadLayer; no wind file → calm conditions.	High
NF R5	VegetationType enum ordinal order is frozen. Constants must only be appended, never reordered, as ordinals are emitted to the frontend for label mapping.	High
NF R6	CaGrid is not thread-safe. The Monte Carlo runner must call deepCopy() before each parallel task — this contract must not be violated.	High
NF R7	All Windows file handles to GeoTIFF files must be released. Coverage must be disposed before reader to avoid Windows lock errors on refresh cycles.	Medium
NF R8	Run history JSON files must be human-readable and not require a database for access, enabling audit without tooling.	Medium

Table 3: Non-Functional Requirements

Functional Requirements

ID	Functional Requirement	Source Package
FR 1	Initialise a 100 m CA grid from ESA WorldCover, Sentinel-2 CV GeoTIFF, ERA5 wind, and OSM road GeoJSON at startup and on each CV poll cycle.	ingestion, grid, facade
FR 2	Run Phase 1 Monte Carlo ensemble (N=200 default) and return ignition probability and damage potential maps as flat float arrays of length rows×cols.	montecarlo, output
FR 3	Run Phase 2 active fire spread using Rothermel ROS-embedded CA transition probabilities and return timestamped GeoJSON fire perimeter snapshots.	simulation, output

FR 4	Accept a CV observation correction payload and apply confirmed burned, suppressed zone, and moisture refresh corrections to the live grid in a fixed order.	correction, cvintegration
FR 5	Persist completed run records as flat JSON files in data/runs/ and serve run history via two REST endpoints.	history, api
FR 6	Detect session mode (PRE_FIRE / ACTIVE_FIRE) based on CV fire perimeter availability and expose it in the session status response.	cvintegration, facade
FR 7	Expose a CORS-enabled REST API permitting GET and POST requests from localhost:3000 and localhost:5173.	config, api
FR 8	Support manual ignition mode: seed Phase 2 fire from a user-supplied GeoJSON polygon when manualIgnition=true in the run request.	facade, ingestion

Table 4: Functional Requirements

4.1.4 Use Cases

Two primary actors interact with WRaP: the Forest Officer (human, via browser dashboard) and the CV Module (automated, via HTTP).

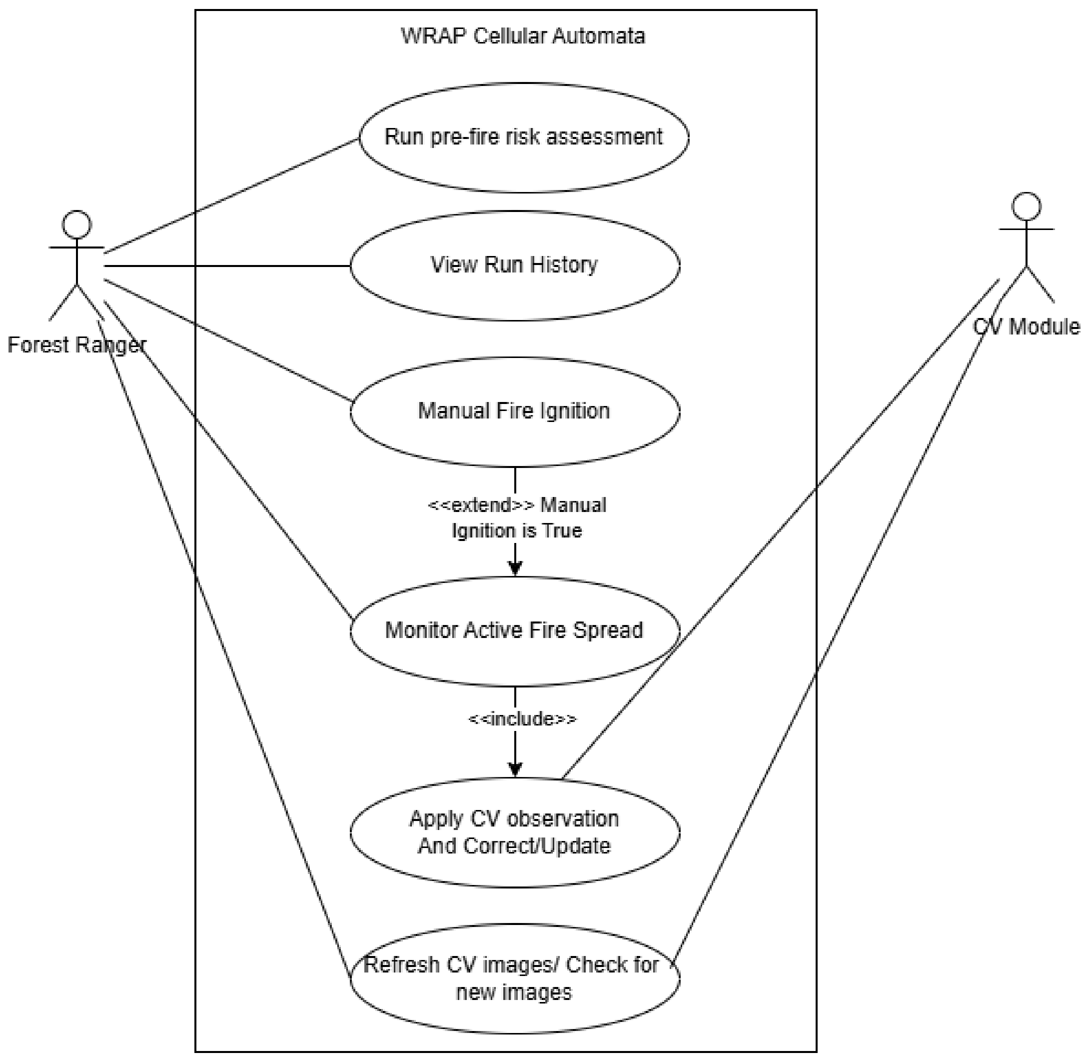

Figure 11: UML Use Case Diagram: actors and system use cases.

4.2 System Design

4.2.1 Architectural overview

WRaP is structured as ten significant packages in a strict layered architecture. Dependency direction is enforced as unidirectional: the API and facade layers depend on mid-level simulation packages, which in turn depend on lower-level domain and physics packages. No lower-level package imports from a higher-level one. This structure was codified as package contract documents before implementation began, enabling group-parallel development without interface ambiguity.

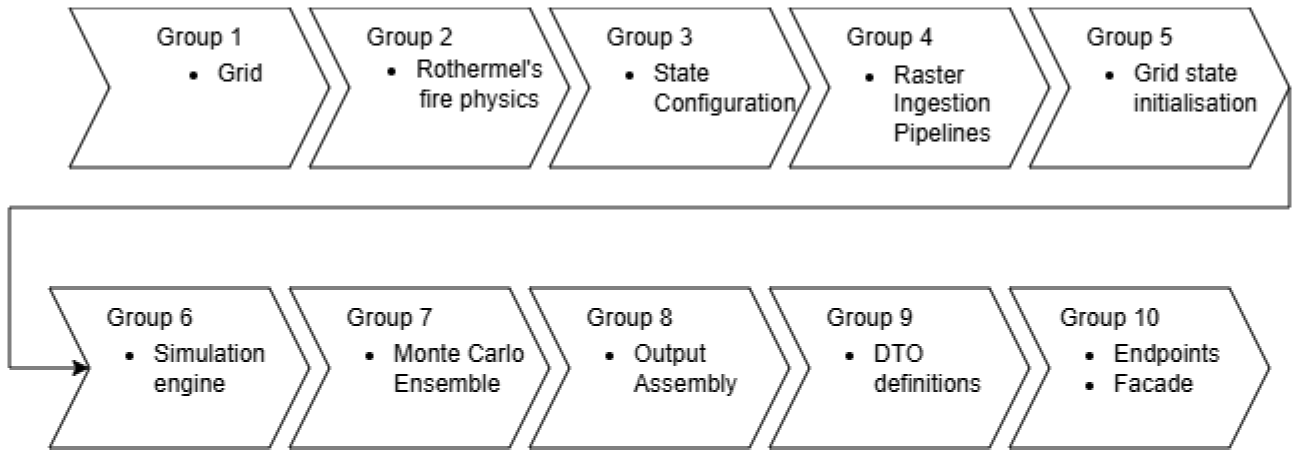

Figure 12: Architecture diagram showing unidirectional package dependencies

This implementation is what enables the overarching architecture that supports phase 1 and phase 2 computation with very high code reuse. For instance, group 7's Monte Carlo Ensemble benefits from the already established fire simulation at group 6. When viewed at a high level, here is the architecture:

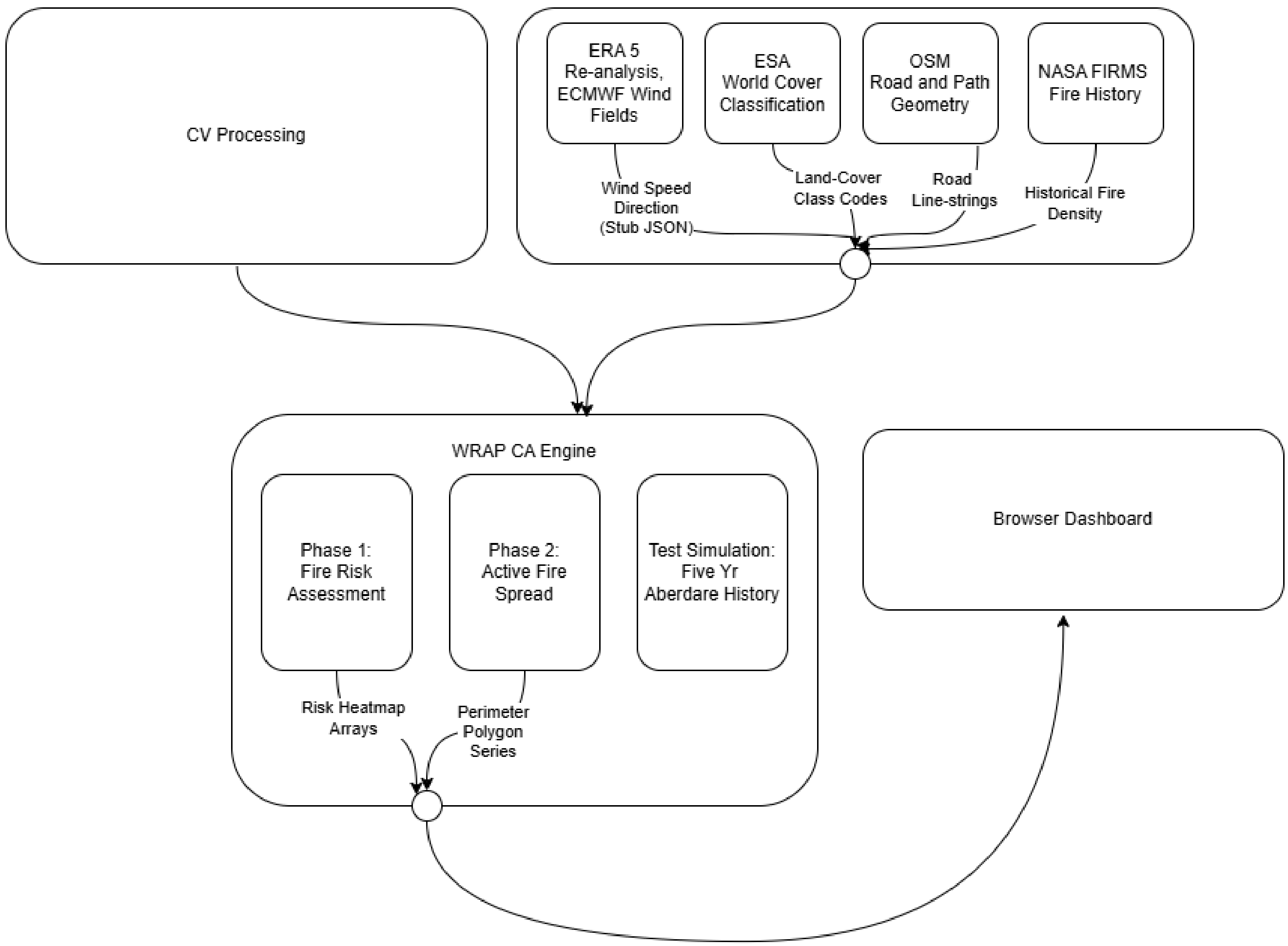

Figure 13: Context diagram of WRaP CA engine modules and external dependencies

4.2.2 CA Grid Domain

The Ca Grid is the simulation's entire spatial state using ordinal values. It holds two parallel 2D *arrays (int[][] states and CellEnvironment[][] environment)* indexed [row][col] with row 0 at the north. A third pair of scalar fields, rows and cols, defines the grid extent. A double cellSizeMetres records the cell side length.

Four discrete states are defined such that an ordinal represents a state

Ordinal	Constant	Meaning	Transition Rules
0	UNBURNED	Vegetated; not yet ignited; eligible to receive fire.	Can transition to BURNING (stochastic ignition) or NON_COMBUSTIBLE (CV correction).
1	BURNING	Currently on fire; source of spread to Moore neighbours.	Transitions to BURNED at the end of each generation. Cannot re-ignite.
2	BURNED	Combustion complete; no remaining fuel.	Absorbing state within a single run. Cannot transition further.
3	NON_COMBUSTIBLE	Permanently inert (water, built area, suppressed zone).	Absorbing state. Applied at init (ESA codes 50/70/80/90/95) or by CV correction.

Table 5: CA State Table

In practice, these four states interact and transition as demonstrated by the finite state machine demonstrated

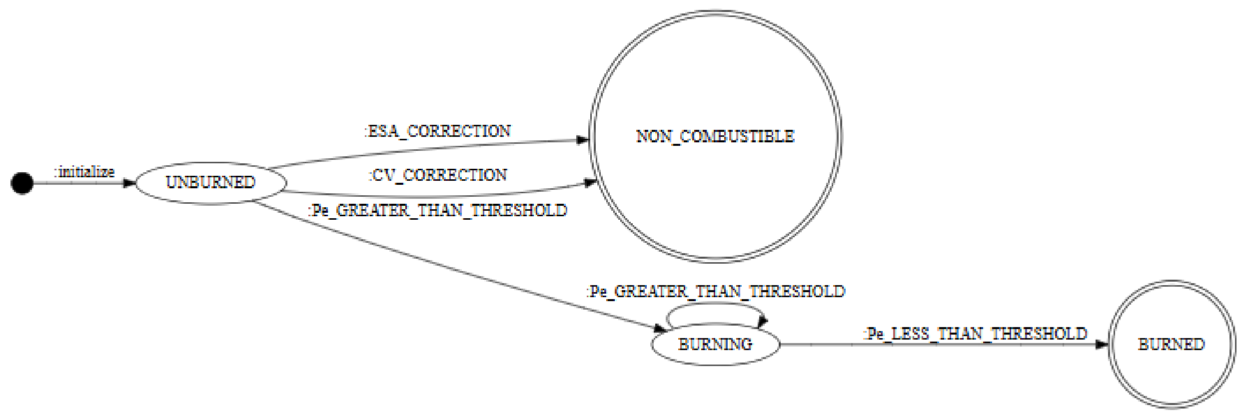

Figure 14: CA finite state machine

4.2.3 Data Ingestion Pipeline

The ingestion pipeline transforms four geospatial input sources into the data objects consumed by the CA grid initialiser. All inputs are read at native resolution and resampled to the CA target resolution before grid construction. Downstream classes have no knowledge that inputs ever existed at a different resolution.

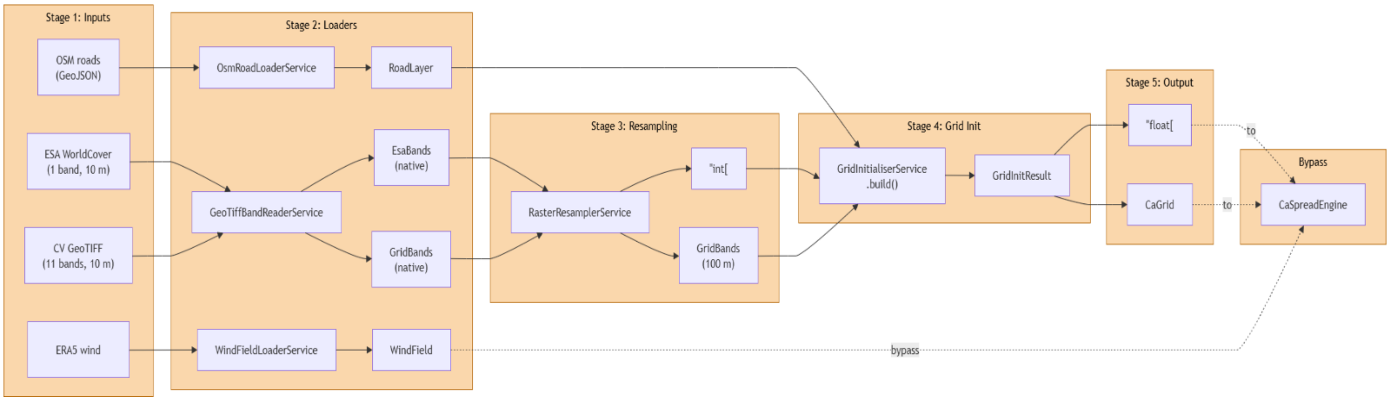

Figure 15: End to end geospatial input pipeline

The inputs include:

CV GeoTIFF Band Layout

The CV GeoTIFF contains 11 bands. 5 of the 11 are consumed in WRaP's current implementation

Band Index (0-based)	Content	Unit	Consumed by
5	NDVI	dimensionless [−1, +1]	GeoTiffBandReaderService → CellEnvironment.ndvi
6	NDMI	dimensionless [−1, +1]	GeoTiffBandReaderService → NDMI scaling → CellEnvironment.ndmi
8	Elevation	metres	GeoTiffBandReaderService → CellEnvironment.elevationMetres
9	Slope	degrees	GeoTiffBandReaderService → Math.toRadians() → CellEnvironment.slopeRadians
10	Aspect	radians [0, 2π]	GeoTiffBandReaderService → CellEnvironment.aspectRadians (passed through)
0–4, 7, 11	Other bands	—	Not read. BandLayout.EXPECTED_BAND_COUNT = 11 enforces file integrity.

Table 6: CV GeoTIFF band layout

ESA WorldCover Class Code Mapping

EsaBandLayout is the single source of truth for mapping ESA class codes to VegetationType constants. This mapping determines which cells are initialised as UNBURNED versus NON_COMBUSTIBLE.

ESA Code	Land Cover Class	VegetationType	Init State
10	Tree Cover	AFROMONTANE_FOREST	UNBURNED
20	Shrubland	SHRUBLAND	UNBURNED
30	Grassland	GRASSLAND	UNBURNED
40	Cropland	CROPLAND	UNBURNED
50	Built-up	BUILT	NON_COMBUSTIBLE
60	Bare / Sparse Veg.	BARE_SOIL	UNBURNED
70	Snow / Ice	BARE_SOIL	NON_COMBUSTIBLE
80	Permanent Water	WATER	NON_COMBUSTIBLE
90	Herbaceous Wetland	WATER	NON_COMBUSTIBLE
95	Mangroves	BARE_SOIL	NON_COMBUSTIBLE
100	Moss / Lichen	BARE_SOIL	UNBURNED
Any other	(unrecognised)	BARE_SOIL	UNBURNED (safe fallback)

Table 7: ESA WorldCover class code to VegetationType mapping

NDMI Scaling

Raw Sentinel-2 NDMI values range from approximately −1 to +1. If stored without transformation, most vegetated cells in a well-watered highland environment would have NDMI above 0, which after scaling maps to moisture fractions above the extinction threshold for GRASSLAND (0.15). This would produce zero rate-of-spread across the majority of the grid.

GridInitialiserService applies a linear scaling inside a private scaledMoisture() method before constructing each CellEnvironment. The mapping has the form:

```
moisture = MOISTURE_MIN + (clamp(rawNdmi, -1, +1) - NDMI_DRY_ANCHOR) *
MOISTURE_SLOPE
moisture = clamp(moisture, MOISTURE_MIN, MOISTURE_MAX)
```

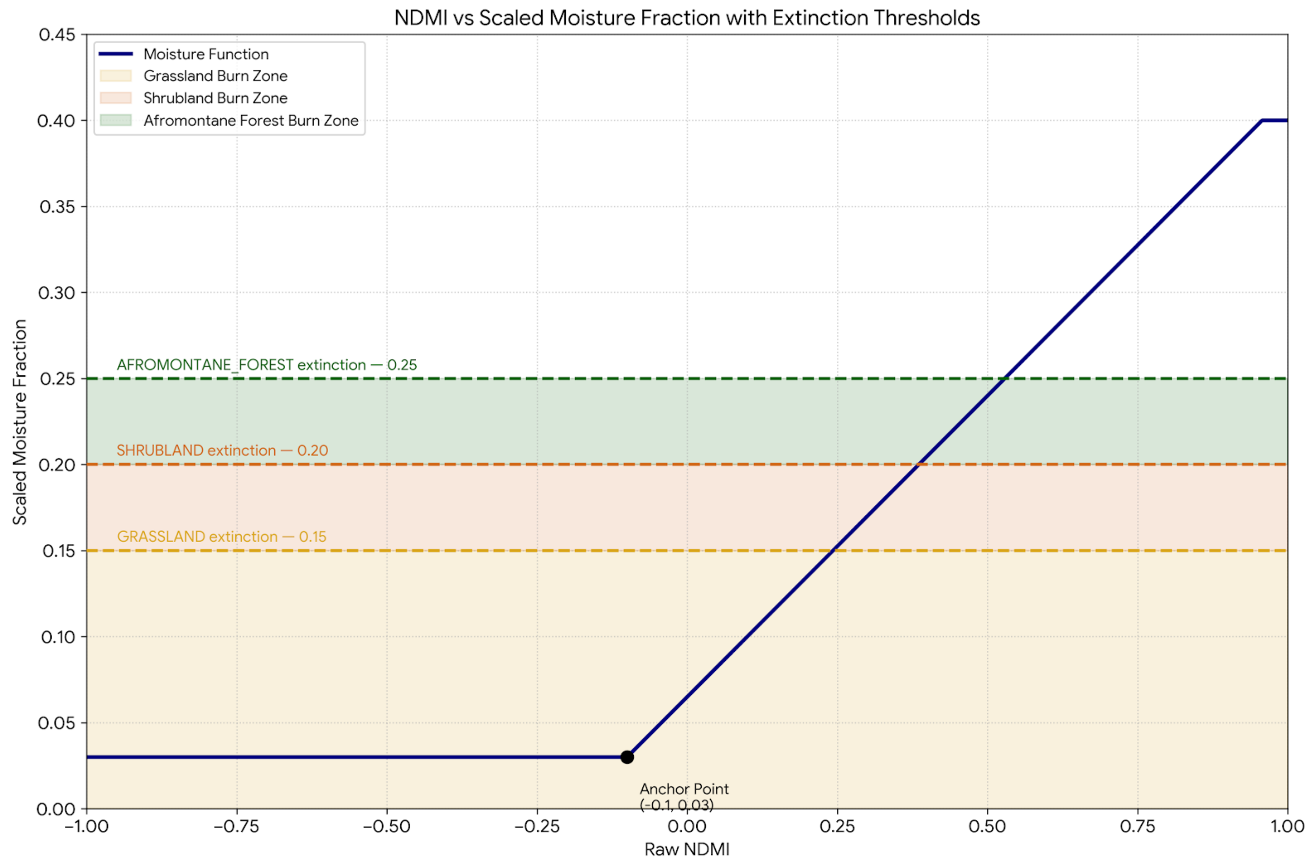

Figure 16: NDMI vs Scaled moisture fraction with extinction thresholds graph

Rothermel Rate of Spread (ROS) Computation

WRaP uses a physics-informed analytical model rather than a trained ML model. The Rothermel ROS pipeline is the analytical core: satellite-derived environmental inputs are transformed into an ignition probability for each cell pair at each generation step.

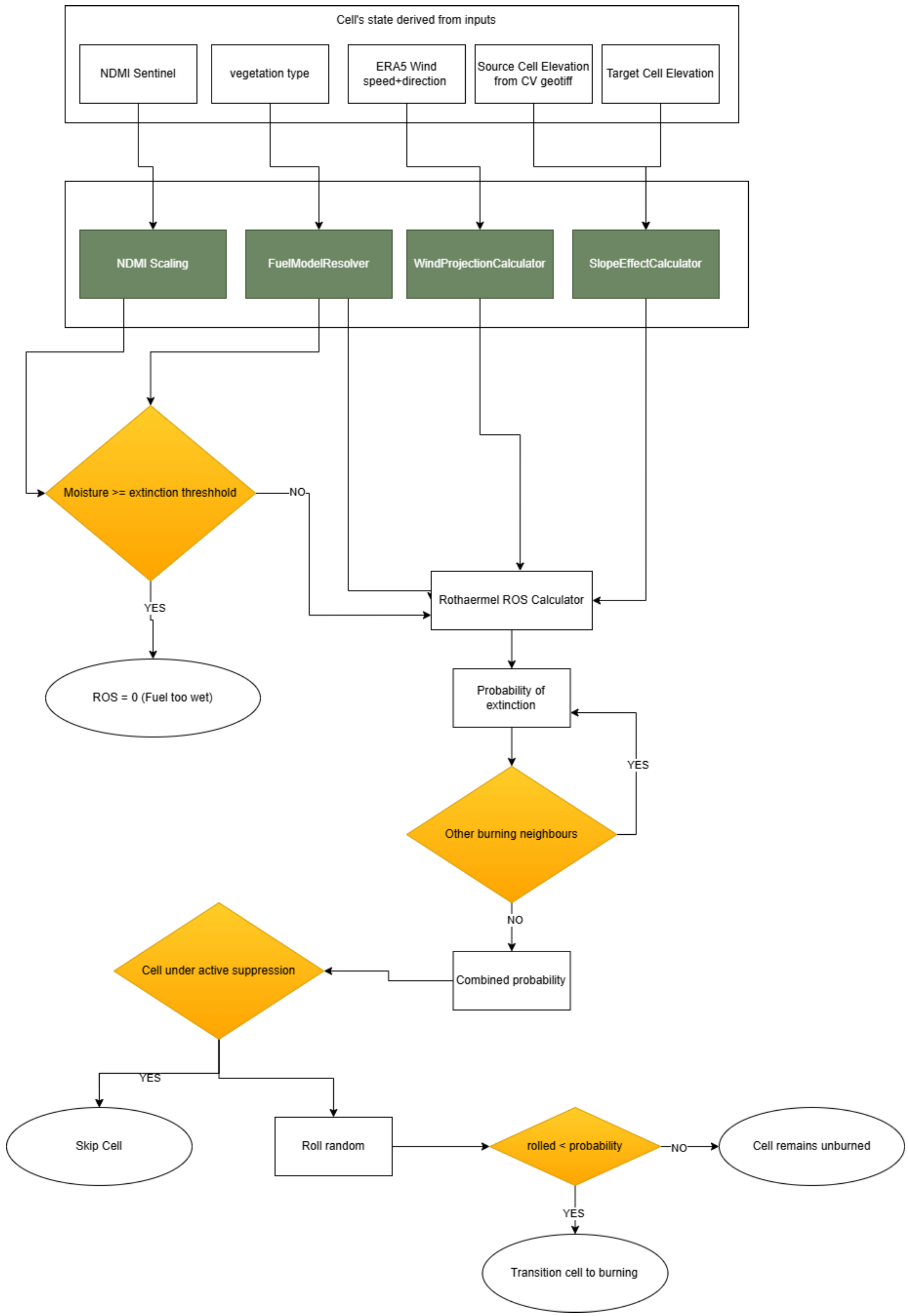

Figure 17: Process Rothermel ROS pipeline Flowchart from satellite inputs to ignition decision.

4.2.4 Phase 1: Monte Carlo Ensemble

Phase 1 answers the pre-fire question: given current fuel conditions across the grid, which cells are most likely to ignite and which areas are most at risk of burning if a fire does start? It does this by running N independent CA spread simulations, each seeded from a different ignition cell, and aggregating the results.

Ignition Likelihood Index — I(c)

The ignition likelihood index combines three per-cell scores into a single weight used for Monte Carlo seeding:

```
I(c) = 0.5 × drynessScore(c)  +  0.2 × firmsScore(c)  +  0.3 × roadScore(c)
```

Term	Weight	Formula	Data Source	Status
Dryness	0.5	1 − clamp(scaledNdmi / 0.40, 0, 1)	CellEnvironment.ndmi (already scaled)	Active
FIRMS density	0.2	normalised historical fire count at cell	FIRMS Kenya dataset	Stubbed to 0.0 (DEV-006)
Road proximity	0.3	1 − (roadDistMetres / maxRoadDistMetres in grid)	float[][] roadProximityMetres	Active

Table 8: Ignition Likelihood Index term weights

NON_COMBUSTIBLE cells always receive weight 0 and cannot be seeded. Combustible cells that score exactly 0 after weighting are floor-clamped to 1e-6, ensuring every unburned cell has a non-zero probability of being selected as an ignition seed — this prevents edge cases where all seeds cluster in one corner of the grid. When *roadProximityMetres[0][0]* equals *Float.MAX_VALUE* (no roads loaded), the road term weight is zeroed out entirely and the remaining weights are applied as-is.

Parallel Execution and Reproducibility

The ensemble runner submits N tasks to a *ForkJoinPool* sized to a thread pool size *(developed and tested on 8 for a 8 physical cores system)*. Each task receives an independent deep copy of the base grid so that concurrent writes to different copies do not interfere. The BurnFrequencyAccumulator is backed by an AtomicIntegerArray of size rows×cols, allowing lock-free concurrent increment operations from all parallel tasks.

Each task receives a unique RNG seeded as new *Random(masterSeed ^ (taskIndex * 0x9e3779b97f4a7c15L))*. The master seed is fixed at 42L to ensure that identical input data produces identical output risk maps, which is important for reproducibility and audit.

4.2.5 Phase 2 Active Fire Spread

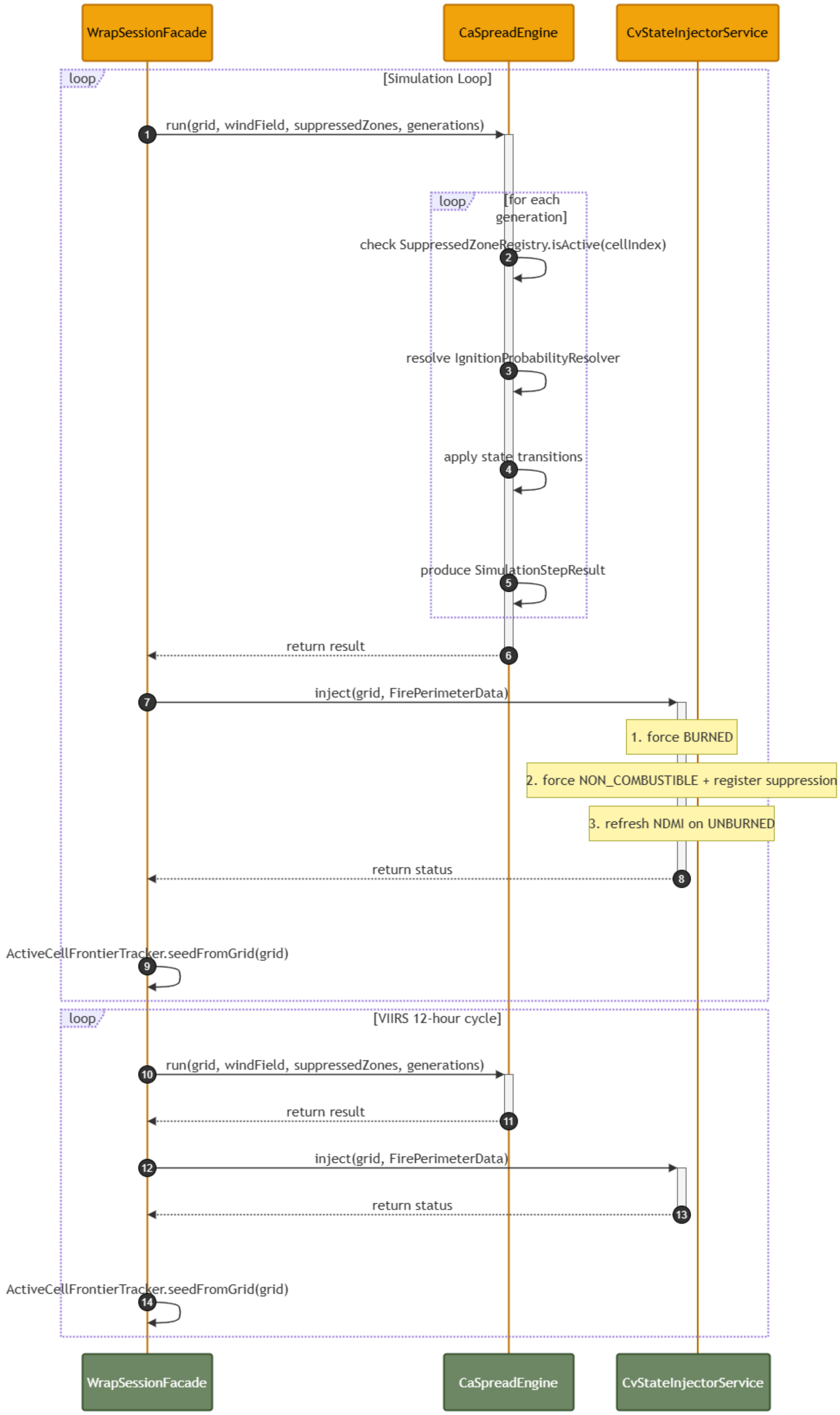

Figure 18: Phase 2 simulation loop sequence diagram

Active Cell Frontier Tracking

CaSpreadEngine only evaluates cells on the active frontier — the set of UNBURNED cells that have at least one BURNING Moore neighbour. This is the core efficiency constraint. Without it, every generation would scan the entire grid, which is O(rows × cols) regardless of how small the fire is. With the frontier tracker, the per-generation cost scales only with the perimeter length of the fire, not the grid size.

Ignition Probability Calculation

For each UNBURNED frontier cell C_i, the engine evaluates all of its BURNING Moore neighbours. For each burning neighbour N_j, it computes a per-neighbour ignition probability Pe_j and then combines them:

```
Ue  = WindProjectionCalculator.effectiveComponent(windSpeed, windDir, (dirIdx +
4) % 8)
phi = SlopeEffectCalculator.slopeAngleRadians(N_j.elevation, C_i.elevation,
dist, isDiagonal)
ROS = RothermelRosCalculator.computeRos(fuel, N_j.ndmi, Ue, phi)
Pe_j = clamp(ROS * timeStepMinutes / distanceMetres, 0.0, 1.0)
P(ignition) = 1 − ∏(1 − Pe_j)  for all BURNING neighbours j
```

The direction index passed to WindProjectionCalculator is *(directionIndex + 4) % 8 — the reverse of the direction used to find the neighbour*. This gives the direction fire would travel from N_j toward C_i, which is the correct direction for wind projection (fire accelerates when wind blows from the burning cell toward the unburned cell).

Suppression Window

The suppression window is fixed at 12 hours inside CvStateInjectorService, matching the approximate VIIRS polar-orbit overpass interval. At the next overpass, the CV module will either re-confirm suppression (resetting the expiry) or produce no entry for the cell, allowing the 12-hour expiry to elapse naturally. Once expired, SuppressedZoneRegistry.isActive() removes the entry lazily on that call and returns false, returning the cell to normal CA evaluation.

4.2.6 REST API Design

Endpoint	Method	Request Body	Response	Notes
/api/session/status	GET	—	SessionStatusResponseDto	Grid summary, mode, past runs. Frontend polls on load.
/api/session/refresh	POST	—	SessionStatusResponseDto	Re-polls CV, re-initialises grid if new data available.
/api/simulation/phase-one/run	POST	PhaseOneRunRequestDto	PhaseOneResultResponseDto	Wind override optional. Returns three float[] arrays + grid dims.
/api/simulation/phase-two/run	POST	PhaseTwoRunRequestDto	PhaseTwoResultResponseDto	simulationHours required. manualIgnition bypasses CV perimeter.
/api/simulation/phase-two/correct	POST	CvCorrectionRequestDto	200 OK (empty body)	Applies correction in-place. Engine must be restarted after.
/api/runs	GET	—	List<RunSummaryResponseDto>	Most-recent-first. Empty list if no runs.
/api/runs/{runId}	GET	—	RunRecord (full detail)	Includes parameters map and result file path.

Table 9 REST API endpoint design

5 Implementation

5.1 WRaP CA Engine

GitHub Repository url: https://github.com/VictorCodebase/wrapca

5.1.1 Project Structure

The project source tree is organised under *com.victorkithinji.wrap.wrapca* and contains thirteen packages mirroring the architectural layers described in Chapter 4.

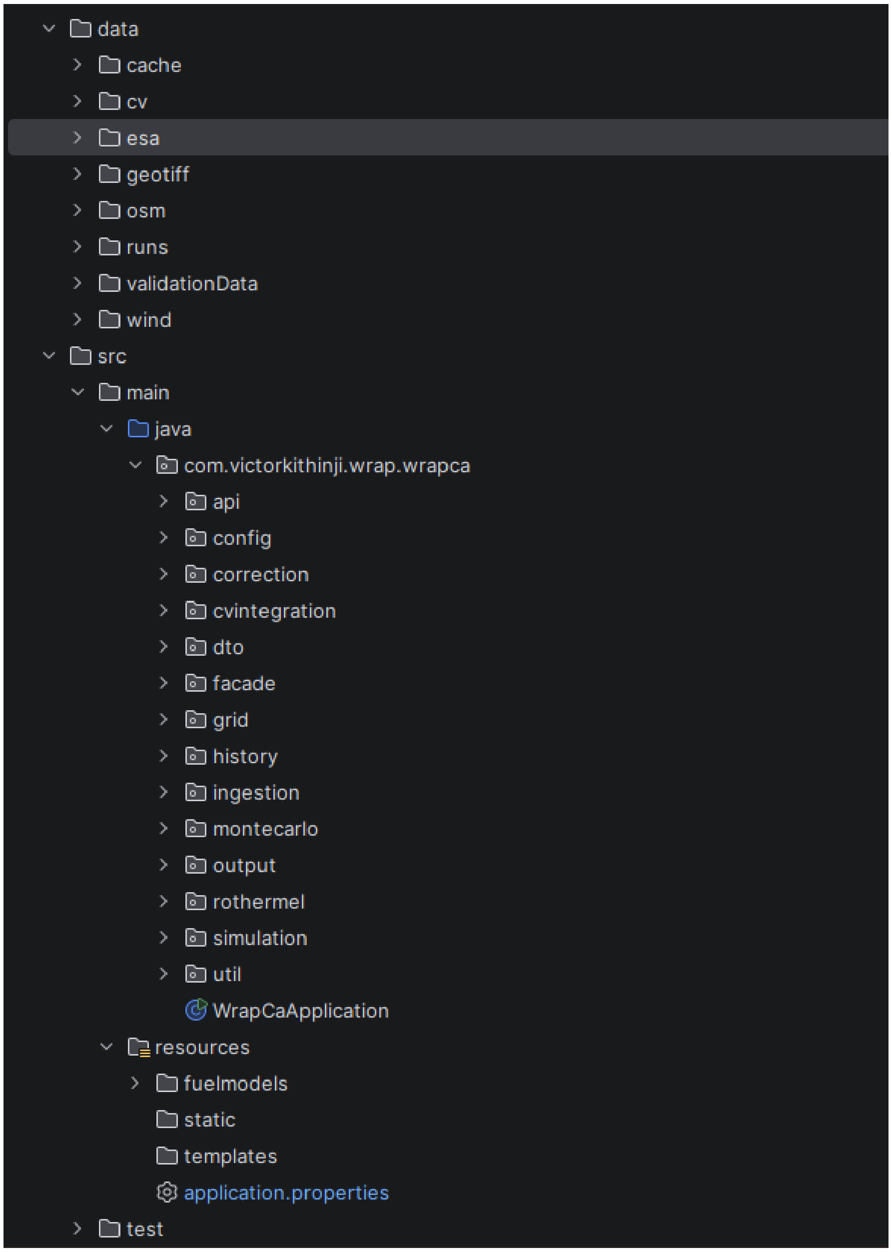

Figure 19: IntelliJ project structure screenshot

5.1.2 CA Engine endpoint response

Phase 1

Endpoint: https://*base_url*/api/simulation/phase-one/run

Body Cookies Headers (8) Test Results

{ } JSON Preview Visualize


```
{
    "runId": "bd8fc187-aa7e-4684-960e-43ed17735042",
    "damagePotentialValues": [ ⋯
    ],
    "ignitionProbabilityValues": [ ⋯
    ],
    "vegetationTypeOrdinals": [ ⋯
    ],
    "rows": 95,
    "cols": 54,
    "analytics": {
        "highRiskCellCount": 5130,
        "highRiskAreaHectares": 513000.0,
        "highRiskAreaByVegetationType": {
            "AFROMONTANE_FOREST": 139100.0,
            "GRASSLAND": 173900.0,
            "SHRUBLAND": 136400.0,
            "CROPLAND": 58000.0,
            "WATER": 2600.0,
            "BUILT": 3000.0
        },
        "topIgnitionSeeds": [
            2784,
            1856,
            1908,
            1909,
            1910
        ],
        "topIgnitionSeedScores": [
            0.03999999910593033,
            0.029999999329447746,
            0.029999999329447746,
            0.029999999329447746,
            0.029999999329447746
        ],
        "dominantVegetationType": "GRASSLAND",
        "simulatedHorizonHours": 24.0,
```


Figure 20: Phase 1 API endpoint response screenshot

Phase 2

Endpoint: [https://*base_url*/api/simulation/phase-two/run](https://base_url/api/simulation/phase-two/run)

Body:

```
{
  "cvDisabled": false,
  "manualIgnition": true,
  "manualIgnitionPolygonGeoJson":
"{\"type\":\"Feature\",\"geometry\":{\"type\":\"Polygon\",\"coordinates\":[[[...]]]},\"properties\":{}}",
  "simulationHours": 3360
}
```

Response:

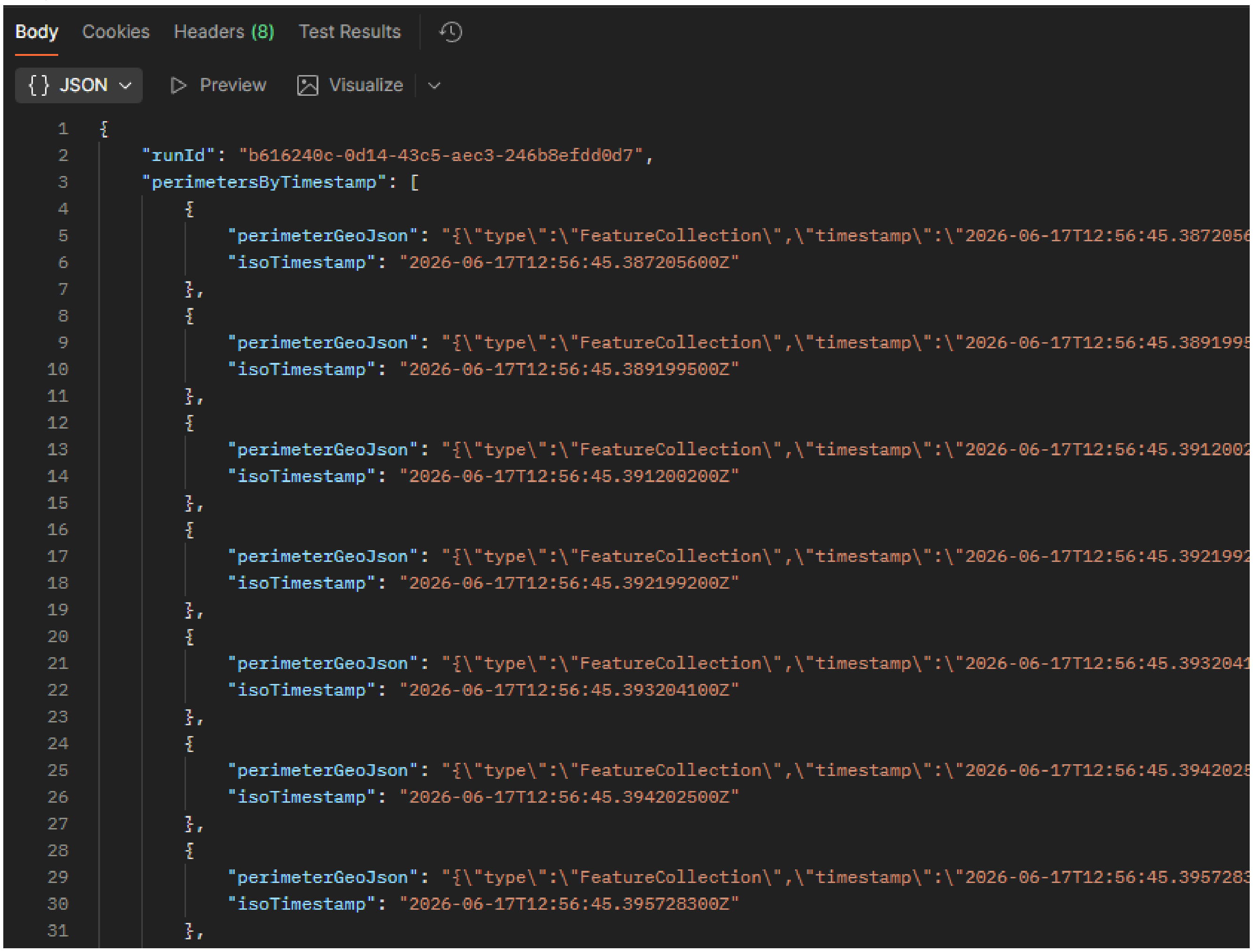

Figure 21: Phase 2 API endpoint response screenshot

5.2 WRaP UI

User inputs and interactions to the WRaP System is done primarily via WRaP UI, a chromium web-based portal. For the sake of this project, a simple browser dashboard is designed enabled by WRaP's Thin Client architecture. WRaP UI to entail primarily canvases and interaction tools.

GitHub repository URL: https://github.com/VictorCodebase/wrap-ui

5.2.1 Phase 1

WRaP UI interface presents CV's Fuel state, and Fuel Risk map rasters which represents the area of interest's state.

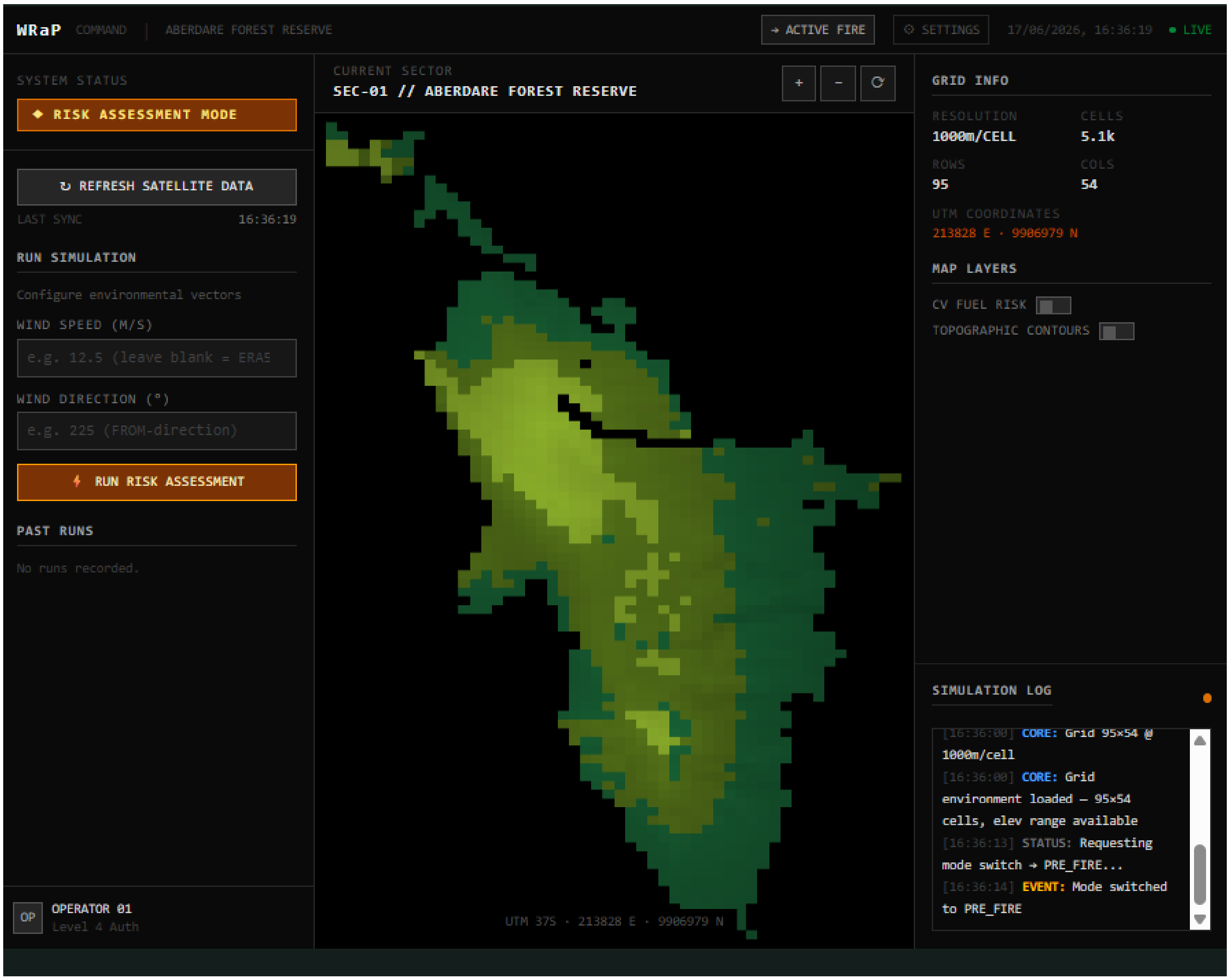

Figure 22: WRaP UI Phase 1 risk assessment interface

The user can trigger a Phase 1 run from here. A successful run provides insights on the top 5 cells of Highest damage frequency, the higher ignition probability *(Ic)* cells and an analysis and average burnt acreage sorted by vegetation type.

This is demonstrated here

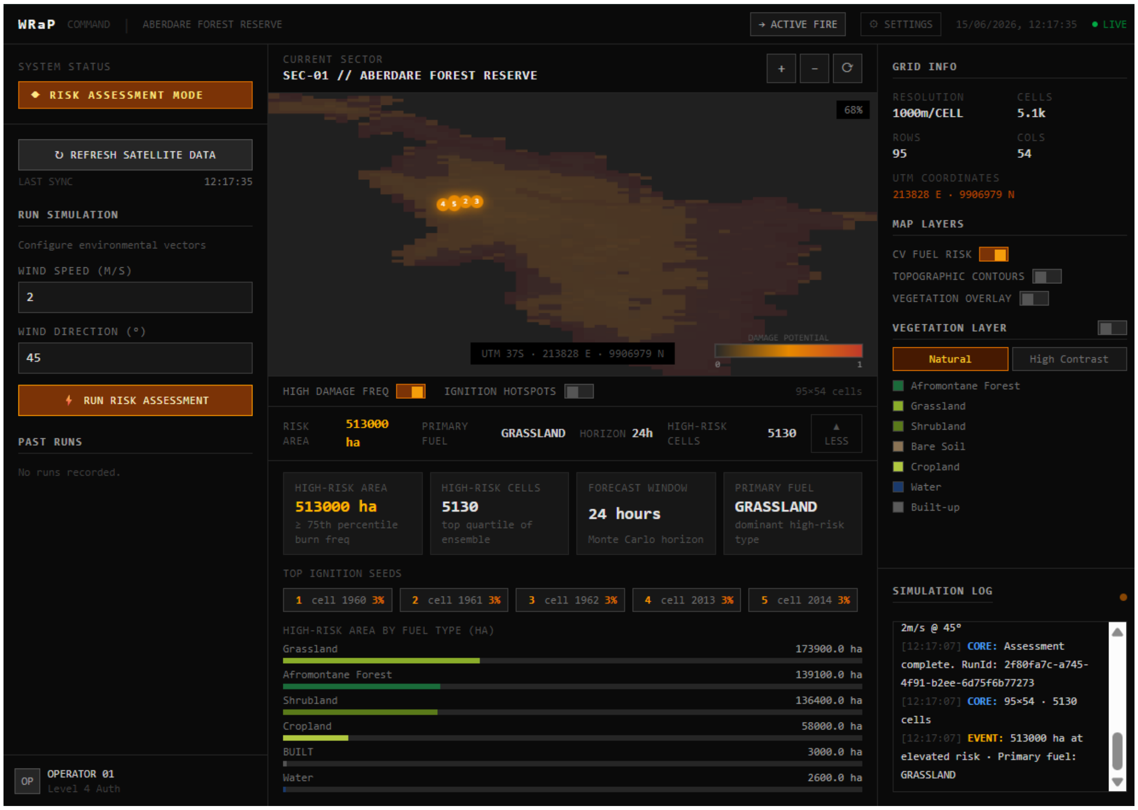

Figure 23: WRaP UI Phase 1 top 5 highest damage frequency cells & burnt acreage by vegetation type

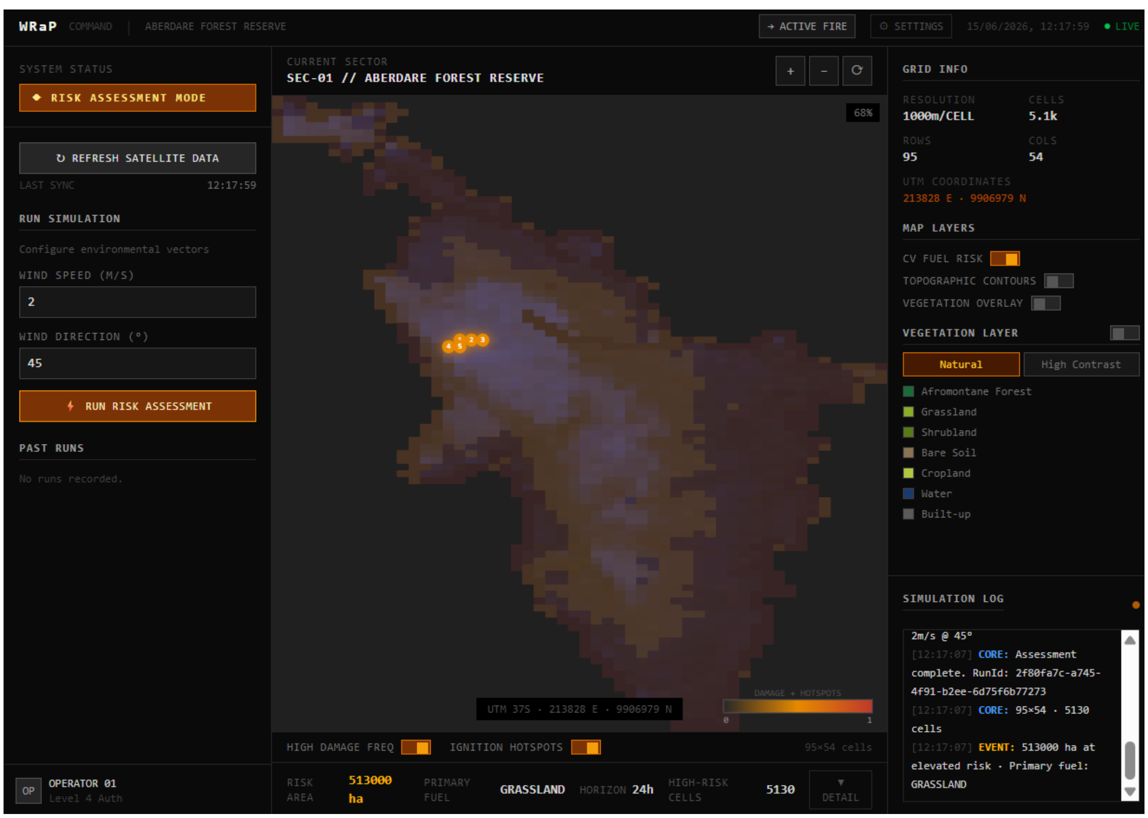

Figure 24: WRaP UI Phase 1 ignition probability (Ic) cells

Phase 2

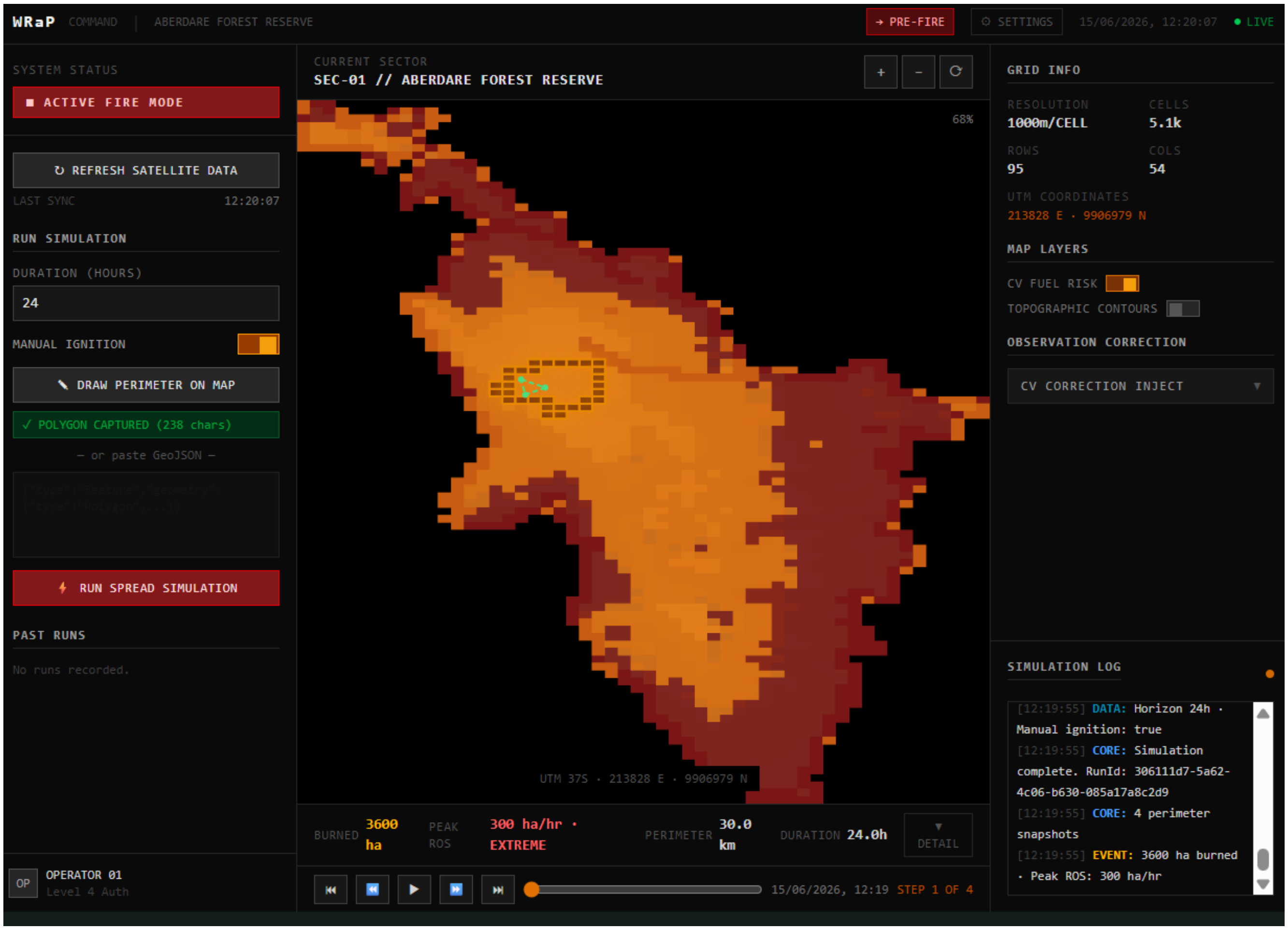

Figure 25: WRaP UI Phase 2 active fire interface

6 Testing, Validation, and Evaluation

6.1 Testing Methodology

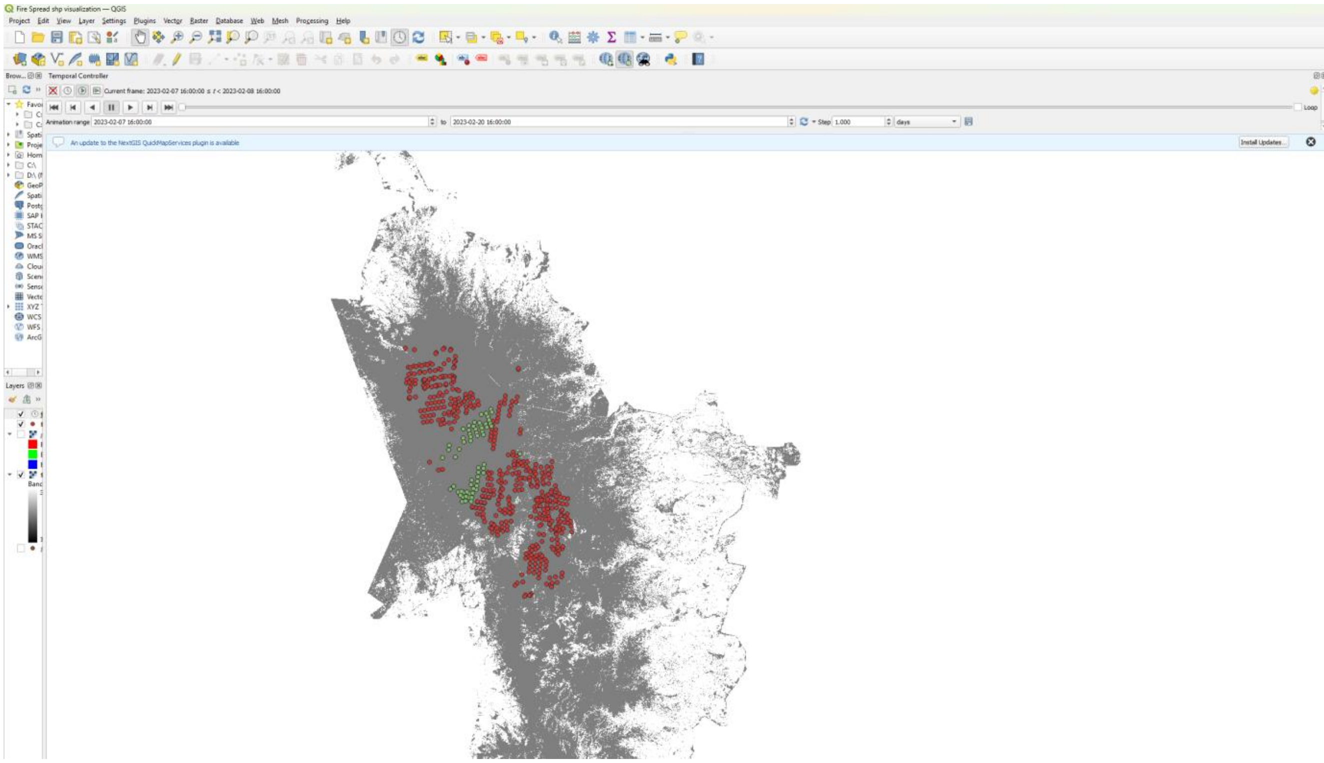

Figure 26: Hotspots captured from the FIRMS VIIRS dataset 2023 February 6th to 22nd

6.1.1 Rothermel Unit Testing

The RothermelRosCalculator was validated against Andrews (2018) reference ROS values before any engine integration. Fixed fuel model inputs, wind speeds, and slope angles were supplied; computed ROS values were asserted within ±5% of published reference values. Edge cases tested include fuel moisture at and above the moisture of extinction threshold (ROS = 0), zero fuel load, and zero effective wind on flat terrain.

6.1.2 CA Spread Integration Testing

A synthetic 50×50 grid with uniform grassland vegetation and known wind and slope inputs was used to verify directional spread preference, frontier tracker correctness, and state transition order. Fire spread preferentially northward (downwind) under a southerly wind as expected. The ActiveCellFrontierTracker set grew and shrank correctly as cells ignited and burned out.

6.1.3 CV Correction Testing

FirePerimeterData objects with known confirmed burned, suppressed zone, and moisture update sets were injected via CvStateInjectorService. Grid state was asserted against expected post-correction values. Suppression expiry was verified by advancing the simulated clock past the 12-hour window and asserting that isActive() returned false for previously suppressed cells.

6.1.4 API Contract Testing

Endpoint response shapes were verified against the DTO contracts. Array lengths were confirmed equal to rows*cols. Ordinal values were confirmed within the valid VegetationType range. HTTP status codes were verified for normal and error conditions.

6.2 Quantitative Validation

The Aberdare National Reserve fire of February 2023 provided the primary validation dataset. VIIRS fire hotspot detections at 24-hour intervals were collected and alpha-shape perimeters were computed from each daily detection set, producing a time series of observed fire perimeters from ignition at coordinates 9960N, 238000E through to 312 hours post-ignition.

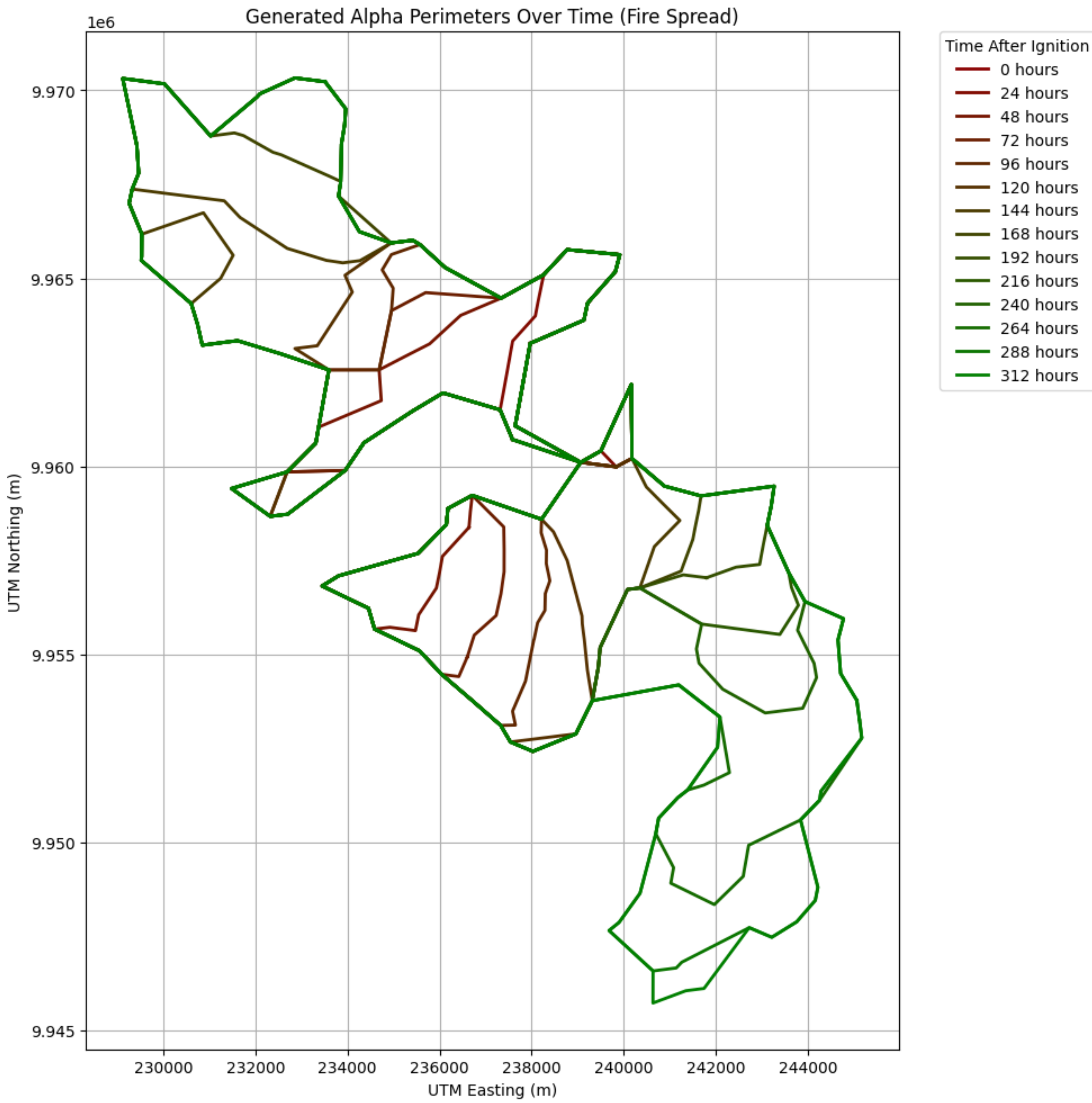

Figure 27: Alpha perimeters from validation dataset (2023 Aberdare fire)

6.2.1 Centroid displacement

Centroid displacement sheds light on any directional bias between the simulated and observed perimeters by capturing the shifting or offset of a cluster's centre-of-mass from its original, expected, or baseline position. Wrap CA here achieves records an average Easting bias delta of +1 cell, and a Northing bias delta of -2 cells.

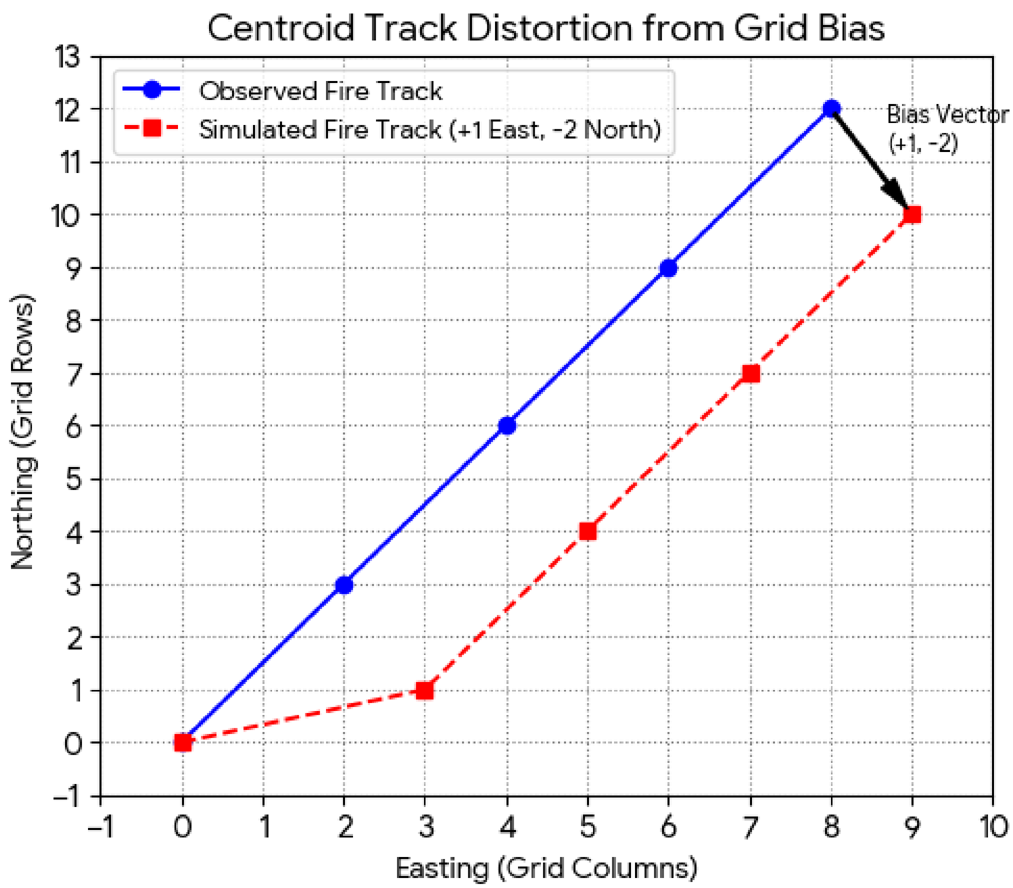

Figure 28: Graph showing general centroid displacement +1E, -2N

The validation shows minimal angular deviation (drift angle) resulting in almost parallel fire tracks (simulated against observed in image above.

6.2.2 Hausdorff Distance

The Hausdorff distance establishes the worst-case boundary deviation in cells against the test data. A score of 1 cell is considered a perfect score. Anywhere up to 5 cells is acceptable, after where results are deemed unreliable. WRAP Cellular Automata modelling achieves a 3-cell deviation from the validation data which is considered a high-accuracy result *(Because Hausdorff measures the absolute worst case)*
Using a cell size of 100m means the simulated fire boundaries will always be expected to be within 300m of the actual fire boundary

6.3 Comparison with Existing System

WRaP occupies a distinct niche relative to existing fire modelling systems. PROPAGATOR (Mediterranean) and Cell2Fire (North American/Canadian FBP) are calibrated for ecosystems and fuel types substantially different from East African Afromontane and savannah environments.
FARSITE is vector-based and scales poorly for large, prolonged fires where grid-based CA methods are more appropriate. FLARE's PHOENIX RapidFire system is the closest state-of-the-art comparator in terms of its CV-CA coupling approach, but operates in a three-dimensional framework requiring substantially more computational resources.

WRaP is differentiated by:

(1) its regional specificity to East African ecosystems;

(2) its two-dimensional CA approach that runs on modest hardware;

(3) its physics-informed rather than historically-trained modelling, avoiding data-dependency issues; and

(4) its closed-loop CV correction mechanism that maintains simulation accuracy across multi-day events.

7. Conclusion

7.1 Summary of Achievements and Contributions

This project has produced WRaP, a fully implemented and validated dual-phase wildfire risk and progression modelling system for East African forest ecosystems. The principal achievements are:

- A complete twelve-package Spring Boot REST API implementing a Rothermel-embedded CA fire spread engine from first principles, with no reliance on proprietary fire modelling libraries
- A pre-fire Phase 1 Monte Carlo ensemble producing dual-layer ignition probability and damage potential risk maps, parallelised across a ForkJoinPool with reproducible seeding.
- A Phase 2 active fire simulation with a 12-hour CV observation correction loop, preventing compounding drift errors across multi-day fire events — the primary gap identified in existing East African fire management tools.
- A validated performance result of 3-cell Hausdorff deviation against the 2023 Aberdare National Reserve fire event, demonstrating physically plausible spread behaviour in the target ecosystem.
- A browser-based thin-client dashboard enabling forest officers to run pre-fire risk assessments, animate fire spread progressions, and inject manual corrections without requiring technical computing expertise.

7.2 Limitations

- Two-Dimensional Only: WRaP does not model spotting, crown fire, or the vertical transport of firebrands. Large-scale fire events in which spotting is a significant spread mechanism will be underestimated by the model.
- ERA5 Wind Resolution: ERA5 provides wind at approximately 9 km horizontal resolution, which is coarser than the 100 m CA grid. Wind is currently applied as a per-cell uniform field from the stub loader; topographic channelling of wind is not captured.

- Static CV Correction Timing: The correction interval is fixed at the VIIRS overpass cycle of approximately 12 hours. Faster-changing fire conditions, particularly during extreme weather events, cannot be corrected at higher frequency without a different satellite source.

7.3 Recommendations and Future Work

- Ensemble Kalman Filter: Replace the hard CV state override correction with a probabilistic Ensemble Kalman Filter assimilation step, providing formal uncertainty quantification for the fire perimeter forecast.
- Higher-Resolution Wind: Integrate ECMWF HRES 10-day forecast data at higher spatial resolution, or apply a diagnostic wind downscaling model, to capture topographic wind channelling effects.
- Spotting Model: Extend the 2D CA with a probabilistic spotting distance model, enabling simulation of firebrand transport ahead of the main fire front.

7.4 Lessons Learned

The most significant technical lesson learned is the value of thorough testing through development. Making this decision allowed me to understand packages that were introducing inaccuracies and errors by unit testing each individually.

I also learned the value of modular code. The Facade module in my code violated this, and it began to introduce issues immediately in understanding my code and what was breaking. All modules, as a result of this lesson, are incredibly modular with very well-defined contracts. Contracts enabled me to not lose track of what modules were offering, and what they required to offer the output.

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Appendix

The project spanned from March 2026 to the end of May 2026. The milestones achieved through this period are listed below alongside their timelines.

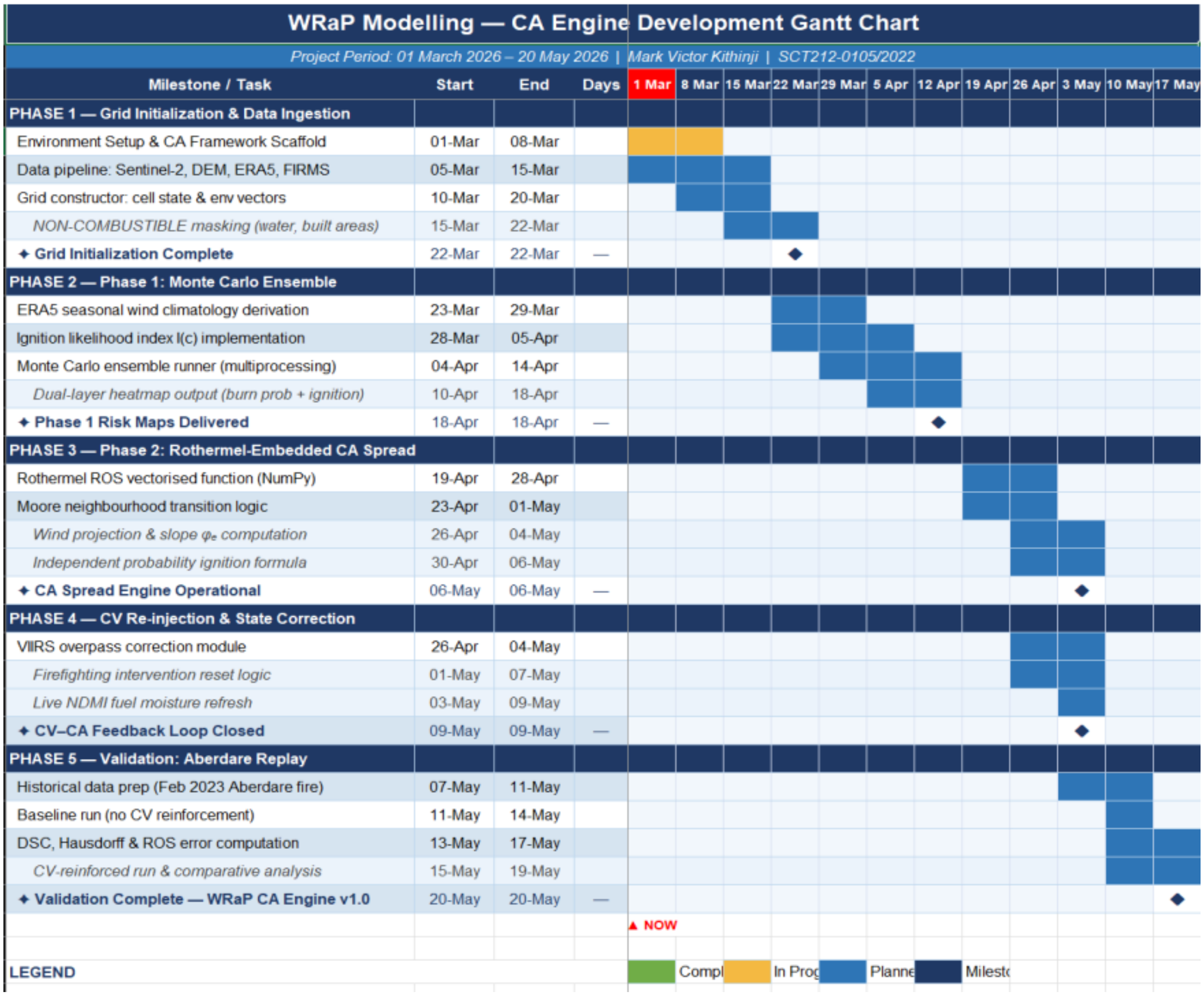

Figure 29: WRaP CA Engine Development Gantt Chart